

New L J Institute of Engineering and Technology, Bodakdev
Branches: CSE (AIML) and CSE (DS)
Semester: II

Subject

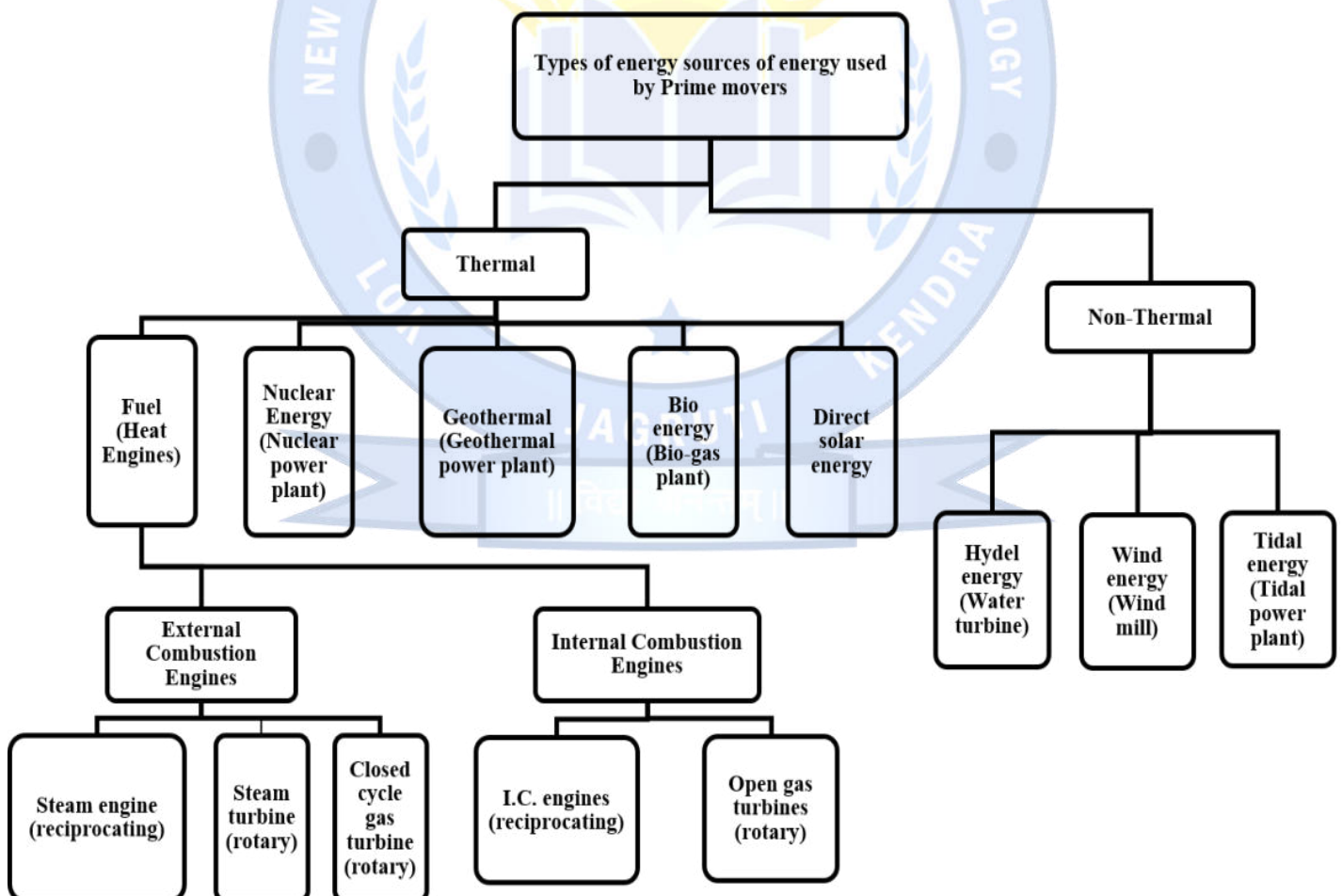
BASIC MECHANICAL ENGINEERING **(BE01000081)**

1. What is Prime mover? How are they classified?

- A prime mover is defined as a device which converts energy from natural sources into mechanical energy or useful work (shaft power).
- Examples of prime movers are: Wind turbine, steam turbine, water turbine, I.C. Engine, etc.

Classification of Prime Movers:

- The prime mover can be classified according to the sources of energy utilized by them.



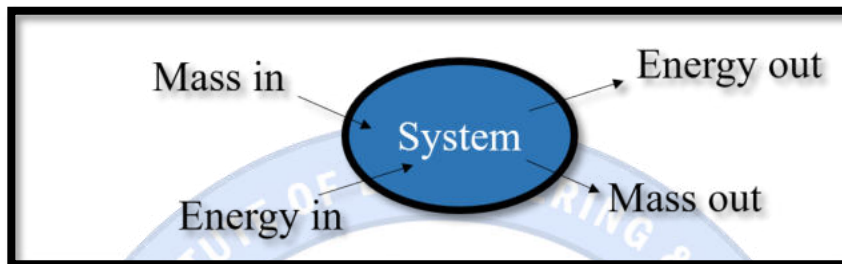
2. Classify and explain the thermodynamic system with giving two suitable engineering examples of each.

- There are three types of thermodynamic system.

(1) Open system (2) Closed system and (3) Isolated system

(1) Open system

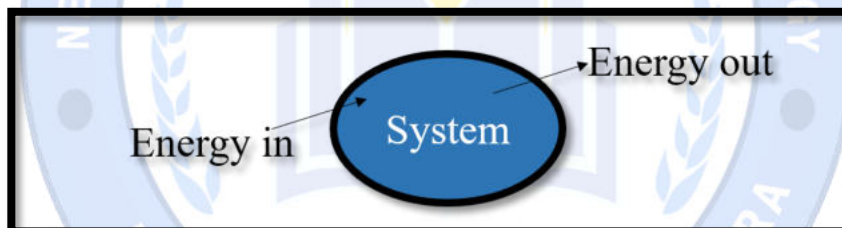
- In an open thermodynamic system, mass and energy (in form of heat and work) both can transfer across the system boundary.
- Such an open system is shown in below figure.



- Examples: Boiler, Turbine, Compressor, Pump, I.C. engine, etc.

(2) Closed system

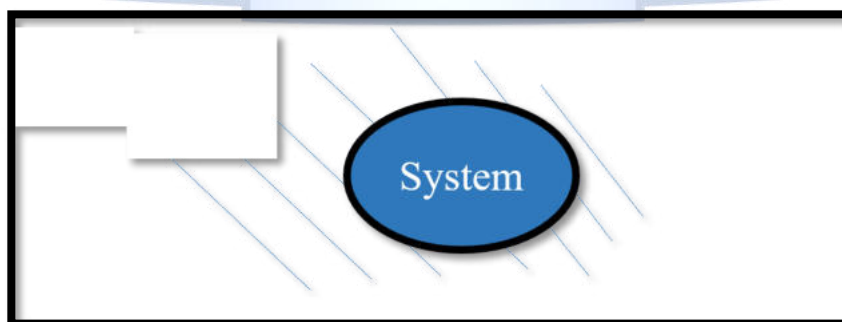
- Closed system is the thermodynamic system, in which only energy transfers take place across the system boundary but there is no mass transfer across boundary.
- Such a closed system is shown in below figure.



- Examples: The gas confined between piston and cylinder, Pressure cooker, Dash Pot, etc.

(3) Isolated system

- Isolated system is a type of thermodynamic system, in which neither mass nor energy transfer takes place across the system boundary.
- Such an isolated system is shown in below figure.



- Examples: The universe; Thermos flask.

3. Explain Specific heat. Give Statements of Zeroth Law and First law of Thermodynamics.

- **Specific Heat:**

- It is defined as the quantity of heat required to raise the temperature of unit mass of the substance by one degree.
- The unit of specific heat is kJ/kg K or J/kg K depending on the unit of Q.
- Mathematically, the heat transfer Q is written as
- $Q = m \times C \times \Delta T$
- $\therefore C = \frac{Q}{m \times \Delta T}$
- Where, C = Specific heat in kJ/kg
m = mass of substance in kg
 ΔT = Temperature difference in K
- The product of mass and specific heat is called heat capacity of substance.
- Specific heat is function of temperature; hence it is not constant but varies with temperature. Generally, it is assumed that it is constant
- The solids and liquids have only one value of specific heat but a gas is considered to have two distinct values of specific heat capacity.
(i) A value when the gas is heated at constant volume, C_v
(ii) A value when the gas is heated at constant pressure C_p
- **The specific heat at constant volume (C_v):** It is defined as the heat required to increase the temperature of the unit mass of a gas by one degree as the volume is maintained constant.
- **The specific heat at constant pressure (C_p):** It is defined as the heat required to increase the temperature of the unit mass of a gas by one degree as the pressure is maintained constant.

- **Zeroth law of Thermodynamics:**

- Zeroth law of Thermodynamics states that if the two bodies, Body A and Body B are in thermal equilibrium with a third Body C separately; then the two bodies A and B shall also be in thermal equilibrium with each other.

- **First law of Thermodynamics:**

- The first law of Thermodynamics is the version of law of conservation of energy applied to Thermodynamic system; and it states that “energy can neither be created nor destroyed, it can be converted from one form to another and total energy remains same, the total energy of an isolated system remains constant”.

4. Prove that the difference between two specific heats of gases is equal to its characteristic gas constant.

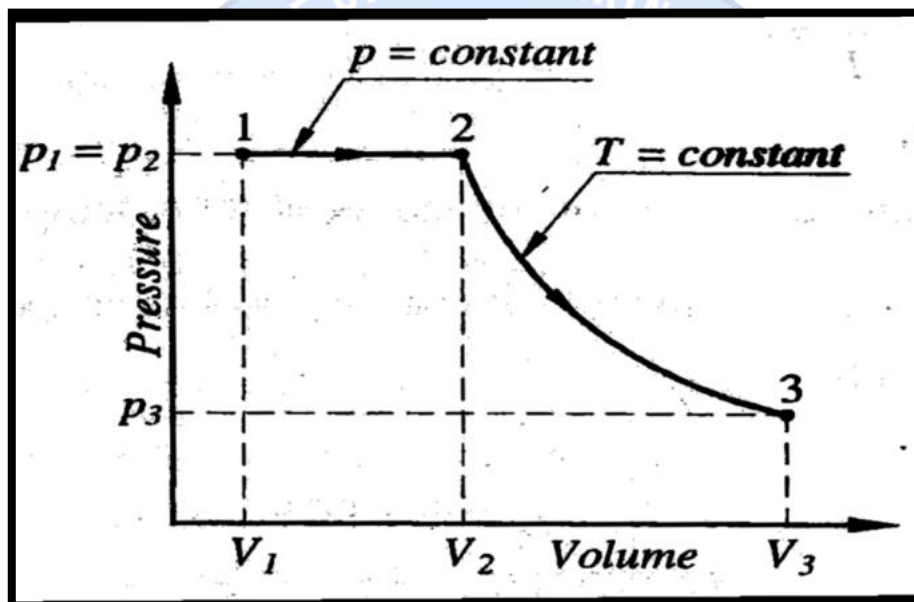
Consider the following notations.

- V_1 and V_2 = Initial and Final Volume of the gas.
 - T_1 and T_2 = Initial and Final Temperature of the gas.
 - p = Pressure of the gas.
 - m = Mass of the gas.
 - C_p and C_v = Specific heat of the gas respectively at constant pressure and at constant volume.
 - R = Characteristic gas constant.
- Consider m kg mass of gas heated at constant volume so as to increase the temperature from T_1 to T_2 .
 - Heat supplied at constant volume,
 $\therefore Q_v = m C_v (T_2 - T_1)$
 - When heat is supplied at constant volume, no external work is done.
 - Thus all heat energy is utilized for changing the internal energy. (According to First law of Thermodynamics)
 $\therefore$ Change in internal energy = $\Delta U = m C_v (T_2 - T_1)$
 - Now consider m kg mass of gas heated at constant pressure so as to increase the temperature from T_1 to T_2 .
 - Heat supplied at constant pressure,
 $\therefore Q_p = m C_p (T_2 - T_1)$
 - When heat is supplied at constant pressure, it is utilized for two purposes.
 1. Increasing the internal energy
 $\therefore$ Increase in internal energy = $m C_v (T_2 - T_1)$
 2. Work done or energy spent in overcoming external resistance.
 $\therefore$ Work done = $p (V_2 - V_1)$
 - Now, Q_p = Increase in internal energy + Work done
 $\therefore Q_p = m C_v (T_2 - T_1) + p (V_2 - V_1)$
 $\therefore m C_p (T_2 - T_1) = m C_v (T_2 - T_1) + p (V_2 - V_1) \dots (1)$
 - Now, let us apply the perfect gas equation to m kg mass of the gas at the beginning and at the end of constant pressure heating.
 $\therefore pV_1 = mRT_1$ and $pV_2 = mRT_2$
 $\therefore p (V_2 - V_1) = mR(T_2 - T_1) \dots (2)$
 - By putting value of eq (2) in eq (1), we get
 - $m C_p (T_2 - T_1) = m C_v (T_2 - T_1) + mR(T_2 - T_1)$
 - After removing common terms on both sides, we get
 - $C_p - C_v = R$ (where C_p is the Specific heat of gas at constant pressure, C_v is the specific heat of gas at constant volume and R is the Characteristic gas constant.)
 - This equation is also known as Mayer's equation.

5. Derive expression $PV/T = \text{constant}$ with the help of Boyle's law and Charles' law.

- In practice, pressure, volume and temperature of a gas changes at the same time.
- As the pressure changes, Charles' law cannot be applied and as the temperature changes, Boyle's law cannot be applied.
- By combining these two laws, the final conditions of the gas can be determined.
- Let us consider a m kg mass of a perfect gas to change its state in the following two successive processes as shown in below p-V diagram
- (i) Process 1-2 at constant pressure, and (ii) Process 2-3 at constant temperature.

- For process 1-2,



- Applying Charles' law,

$$\bullet \quad \frac{V_1}{T_1} = \frac{V_2}{T_2} \quad (\because p = \text{Constant})$$

$$\therefore \frac{V_1}{T_1} = \frac{V_2}{T_3} \quad (\because T_2 = T_3)$$

$$\therefore V_2 = \frac{T_3 V_1}{T_1} \quad \dots (1)$$

- For process 2-3,

- Applying Boyle's law,

$$\bullet \quad p_2 V_2 = p_3 V_3 \quad (\because T = \text{Constant})$$

$$\therefore p_1 V_2 = p_3 V_3 \quad (\because p_2 = p_1)$$

$$\therefore V_2 = \frac{p_3 V_3}{p_1} \quad \dots (2)$$

- By comparing eq. (2) and eq. (3)

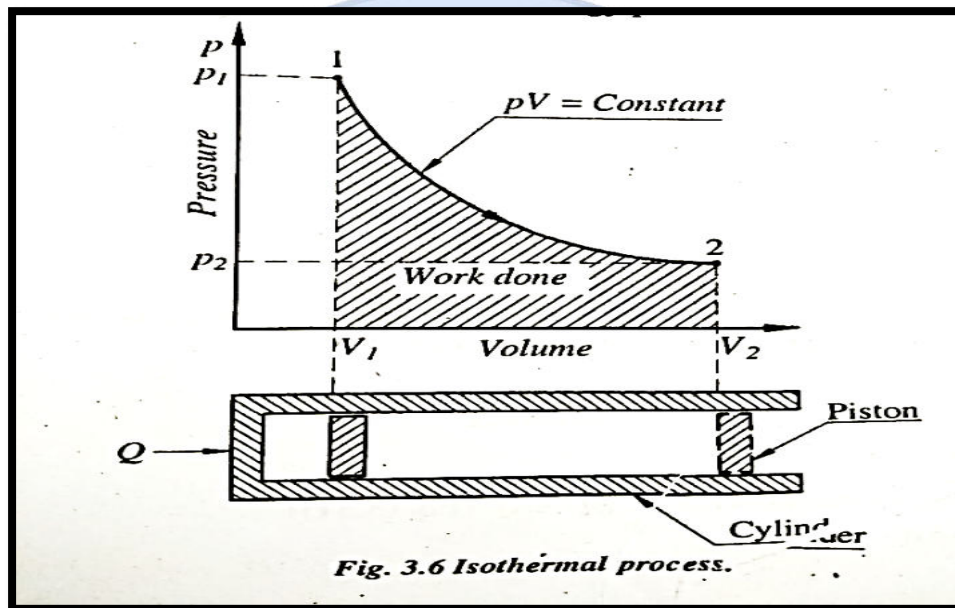
$$\therefore \frac{T_3 V_1}{T_1} = \frac{p_3 V_3}{p_1}$$

$$\therefore \frac{p_1 V_1}{T_1} = \frac{p_3 V_3}{T_3}$$

$$\therefore \frac{pV}{T} = \text{constant}$$

6. Describe Isothermal process and derive expression for Work done, Change in Internal energy, Heat transfer and Change in Enthalpy.

- In an isothermal process, the temperature of the gas remains constant during the process.
- This process follows Boyle's law.



- Thus, the law of expansion or compression for isothermal process on p – V diagram is hyperbolic as p is inversely varies as V.
- Thus, this process is also known as hyperbolic process or Constant Internal energy process.
- **Work done during the process**

- Work done, $W = \text{Area below the curve}$

$$\therefore W = \int_{V_1}^{V_2} p \, dV \dots (1)$$

But for isothermal process,

$pV = C$ (According to Boyle's Law)

$$\therefore p = C/V$$

Substituting value of p in eq (1)

$$\therefore W = \int_{V_1}^{V_2} \frac{C}{V} \, dV$$

$$\therefore W = C \int_{V_1}^{V_2} \frac{1}{V} \, dV$$

$$\therefore W = C \ln \frac{V_2}{V_1}$$

$$\therefore W = C \ln \frac{V_2}{V_1} \quad (\because pV = p_1 V_1 = C)$$

$$\therefore W = p_1 V_1 \ln \frac{V_2}{V_1}$$

$$\therefore W = p_1 V_1 \ln r$$

- Where, $\frac{V_2}{V_1} = \text{expansion ratio} = r$

- But, $pV_1 = mRT_1$ and $pV_2 = mRT_2$

$$\therefore W = mRT_1 \ln r = mRT_2 \ln r$$

- Change in internal energy**

- Change in internal energy $= \Delta U = m C_v (T_2 - T_1)$

But, $T_2 = T_1$ (since it is Isothermal Process)

$$\therefore \Delta U = 0$$

So, during the isothermal operation, change in internal energy is zero and hence this process is also known as constant internal energy process.

- Heat Transfer:**

- According to first law of thermodynamics:

- $\delta Q = \delta W + dU$

- But, $dU = 0$

- $\therefore \delta Q = \delta W$

$$\therefore \delta Q = p_1 V_1 \ln \frac{V_2}{V_1} = p_1 V_1 \ln \frac{P_1}{P_2} = mRT_1 \log_e \frac{V_2}{V_1}$$

- Change in enthalpy**

- Change in enthalpy $= \Delta H = m C_p (T_2 - T_1)$

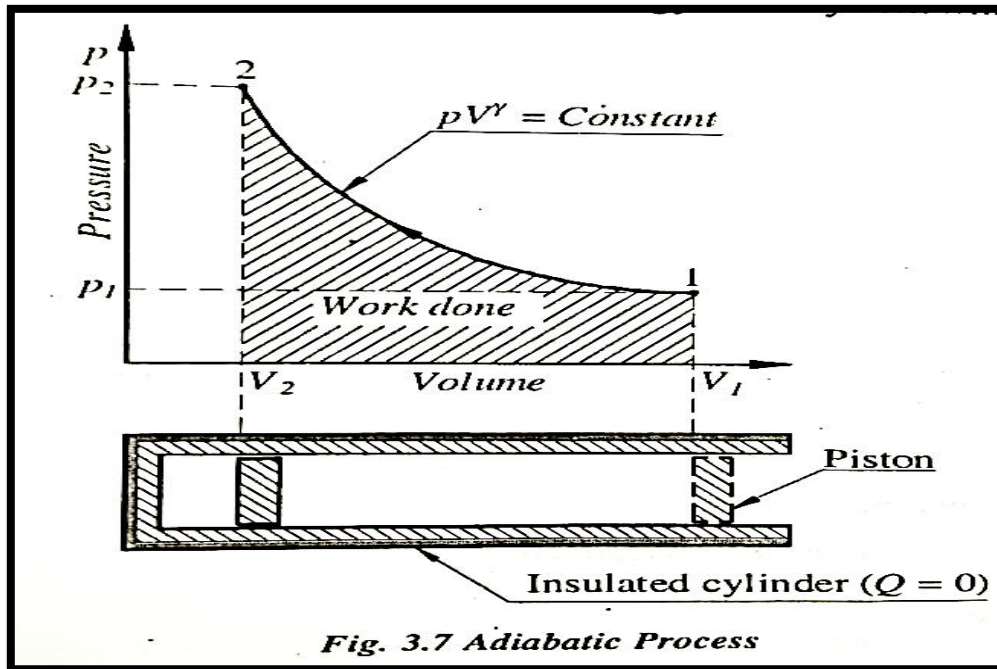
But, $T_2 = T_1$ (since it is Isothermal Process)

$$\therefore \Delta H = 0$$

7. What is adiabatic process? Prove with usual notations the law of governing adiabatic process as $PV^\gamma = \text{Constant}$.

- An adiabatic process is the thermodynamic non-flow process in which there is no heat interaction during the process, i.e. during the process $Q = 0$.
- In this process, the work interaction is there at the expense of internal energy.
- There is no supply of heat takes place during the adiabatic process.
- Frictionless adiabatic process is known as **isentropic process**.

- Let gas contained in piston cylinder arrangement as shown in figure.
- **Law of adiabatic process ($pV^\gamma = \text{Constant}$)**
 - Let us consider unit mass of gas



- Now, according to first law of thermodynamics,
 - Heat supplied = Change in internal energy + work done
- $$\therefore dq = C_v dT + p dv$$
- $$\therefore C_v dT + p dv = 0 \dots (1) \quad (\because \text{For adiabatic process, } dq = 0)$$

Now from characteristic gas equation:

$$pv = RT \quad (\because m = 1 \text{ kg})$$

By differentiating this equation, we get

$$\therefore p dv + v dp = R dT$$

$$\therefore dT = \frac{p dv + v dp}{R}$$

By substituting value of dT in eq (1), we get

$$\therefore C_v \left(\frac{p dv + v dp}{R} \right) + p dv = 0$$

$$\therefore C_v (p dv + v dp) + (R \times p dv) = 0$$

$$\therefore C_v (p dv + v dp) + (C_p - C_v \times p dv) = 0 \quad (\because C_p - C_v = R)$$

$$\therefore C_v p dv + C_v v dp + C_p p dv - C_v p dv = 0$$

$$\therefore C_v v dp + C_p p dv = 0$$

Now, dividing above equation $C_v \cdot pv$

$$\therefore \frac{dp}{p} + \left(\frac{c_p}{c_v} \times \frac{dv}{v} \right) = 0$$

$$\therefore \frac{dp}{p} + \left(\gamma \times \frac{dv}{v} \right) = 0 \quad \left(\because \frac{c_p}{c_v} = \gamma \right)$$

Now, integrating above equation both sides, we get

$$\therefore \log_e p + \log_e v^\gamma = \log_e C$$

$$\therefore p v^\gamma = \text{Constant}$$

The above equation is also known as **Poisson's law**.

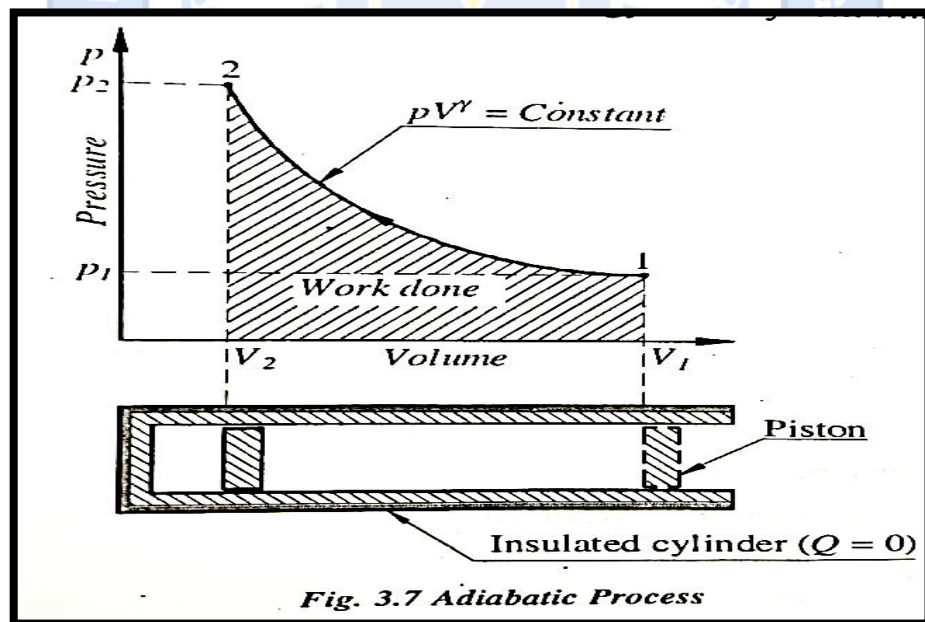
$$\text{Also, specific volume } v = \frac{V}{m}$$

$$\therefore p \cdot \frac{V^\gamma}{m^\gamma} = \text{Constant}$$

$$\therefore p V^\gamma = \text{Constant} \quad (\because m^\gamma \times c = c, \text{ another constant})$$

8. For adiabatic process, derive relation between pressure, volume and temperature. Also derive equation for work done and heat transfer.

- p-V diagram for adiabatic process shown in below figure.



- **Relation between p, V and T**

- We have, $pV^\gamma = C$ (Adiabatic Law) and $\frac{pV}{T} = c$ (Combined gas law)

By dividing above two equations, we get

- $V^{\gamma-1} \times T = \text{constant}$

- so, we can write as,

$$V_1^{\gamma-1} T_1 = \text{constant and } V_2^{\gamma-1} T_2 = \text{constant}$$

- From above equation, we get

$$V_1^{\gamma-1} T_1 = V_2^{\gamma-1} T_2$$

$$\therefore \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \dots (1)$$

- Now, we know $p_1 V_1^\gamma = p_2 V_2^\gamma$

$$\frac{p_1}{p_2} = \left(\frac{V_2}{V_1}\right)^\gamma$$

$$\therefore \frac{V_2}{V_1} = \left(\frac{p_1}{p_2}\right)^{1/\gamma} \dots (2)$$

- From eq(1) and (2), we get

$$\frac{T_1}{T_2} = \left(\frac{p_1}{p_2}\right)^{\gamma-1/\gamma}$$

$$\therefore \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{p_1}{p_2}\right)^{\gamma-1/\gamma}$$

- **Work done during the process**

- Work done, $W = \text{Area below the curve}$

$$\therefore W = \int_{V_1}^{V_2} p \, dV \dots (1)$$

But for adiabatic process, $pV^\gamma = C$

$$\therefore p = C/V^\gamma$$

Substituting value of p in eq (1)

$$\therefore W = \int_{V_1}^{V_2} \frac{C}{V^\gamma} \, dV$$

$$\therefore W = C \int_{V_1}^{V_2} \frac{1}{V^\gamma} \, dV$$

$$\therefore W = C \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2}$$

$$\therefore W = \frac{C}{-\gamma+1} (V_2^{-\gamma+1} - V_1^{-\gamma+1})$$

$$\therefore \text{But, } C = p_1 V_1^\gamma = p_2 V_2^\gamma$$

• **By putting appropriate value of C , we get**

$$\therefore W = \frac{1}{-\gamma+1} (p_2 V_2^\gamma V_2^{-\gamma+1} - p_1 V_1^\gamma V_1^{-\gamma+1})$$

$$\therefore W = \frac{p_1 V_1 - p_2 V_2}{\gamma-1}$$

$$\text{But, } p_1 V_1 = mRT_1 \text{ and } p_2 V_2 = mRT_2$$

$$\therefore W = \frac{mR(T_1 - T_2)}{\gamma - 1}$$

$$\therefore W = \frac{m(C_p - C_v)(T_1 - T_2)}{\gamma - 1} \quad (\because C_p - C_v = R)$$

$$\therefore W = \frac{m(C_p - C_v)(T_1 - T_2) \times C_v}{C_p - C_v} \quad \left(\because \gamma - 1 = \frac{C_p - C_v}{C_v} \right)$$

Also $W = m C_v (T_1 - T_2)$ = change in internal energy

- **Heat Transfer:**

- During adiabatic process, heat transfer is zero.
- $\delta Q = 0$

9. One cubic meter of air at pressure of 1.5 bar and 80 deg C is compressed to final pressure 8 bar and volume 0.28 m³. Determine (i) Mass of air (ii) Index of compression 'n' (iii) Change in internal energy (iv) Heat transfer during compression. Take $\gamma = 1.4$ and $R = 287$ J/kg K.

- **Given Data:**

- Initial volume of air, $V_1 = 1 \text{ m}^3$
- Initial pressure of air, $p_1 = 1.5 \text{ bar} = 1.5 \times 10^5 \text{ Pa}$
- Initial Temperature of air, $T_1 = 80^\circ \text{C} = 353 \text{ K}$
- Final pressure of air, $p_2 = 8 \text{ bar} = 8 \times 10^5 \text{ Pa}$
- Final volume of air, $V_2 = 0.28 \text{ m}^3$
- $\gamma = 1.4$
- $R = 287 \text{ J/kg K}$

- **To Find:**

- Mass of air; m
- Polytropic index; n (thus it is Polytropic process)
- Change in internal energy; ΔU
- Heat transfer; Q

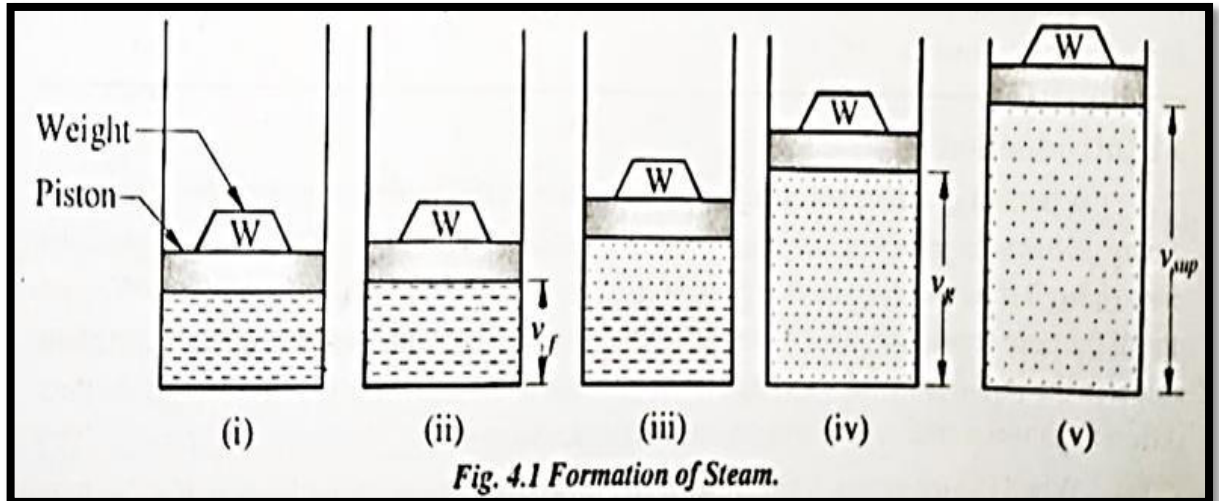
- **Calculations:**

- For mass of air (m):
 - $p_1 V_1 = mRT_1$
 - $m = \frac{p_1 V_1}{RT_1}$
 - $m = \frac{1.5 \times 10^5 \times 1}{287 \times 353}$

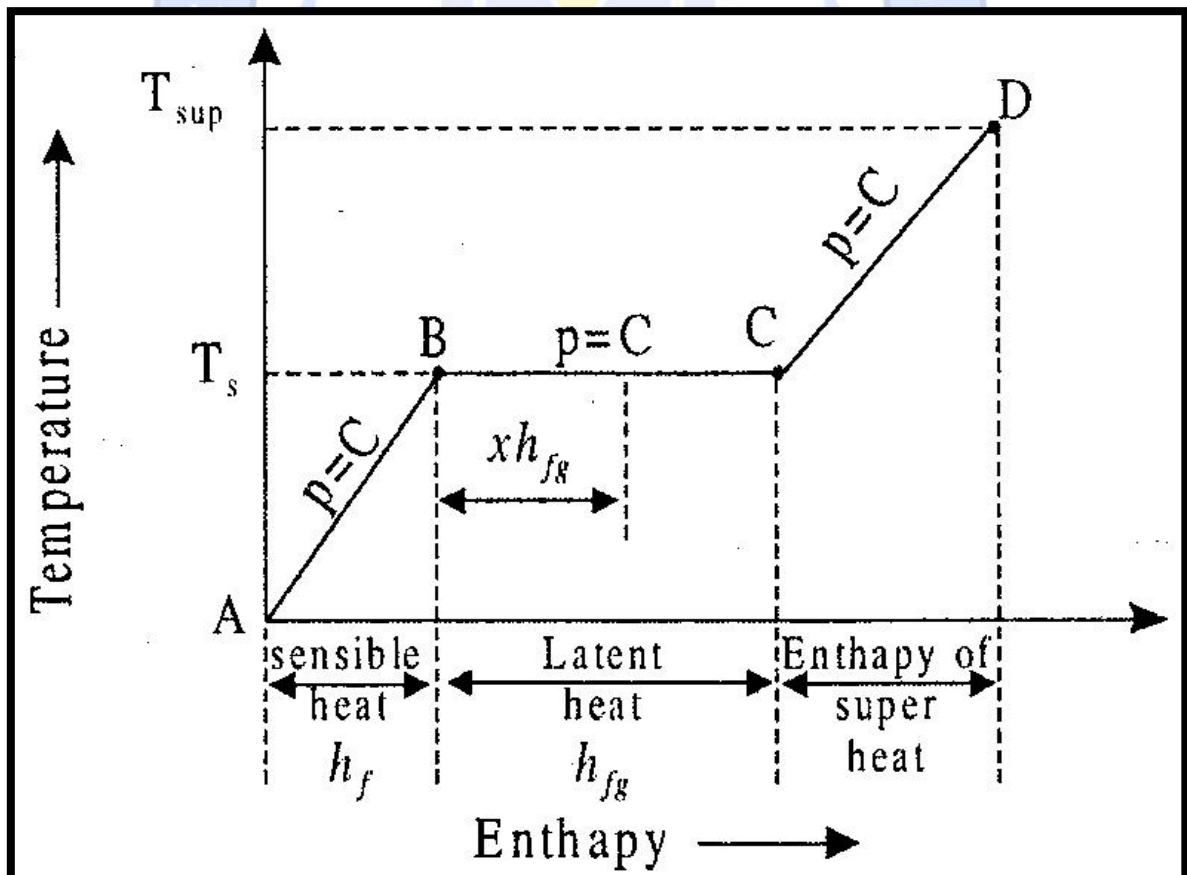
- **$m = 1.4805 \text{ kg}$**
- For Polytropic index (n):
 - $p_1 V_1^n = p_2 V_2^n$
 - $\frac{p_1}{p_2} = \left(\frac{V_2}{V_1}\right)^n$
 - $\ln\left(\frac{p_1}{p_2}\right) = n \ln\left(\frac{V_2}{V_1}\right)$
 - $n = \frac{\ln\left(\frac{p_1}{p_2}\right)}{\ln\left(\frac{V_2}{V_1}\right)}$
 - $n = \frac{\ln\left(\frac{1.5 \times 10^5}{8 \times 10^5}\right)}{\ln\left(\frac{0.28}{1}\right)}$
 - **$n = 1.3118$**
- Change in internal energy (ΔU):
 - $\Delta U = m C_v (T_2 - T_1) \dots (1)$
 - For T_2 ,
 - $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{n-1}$
 - $T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{n-1}$
 - $T_2 = 353 \left(\frac{1}{0.28}\right)^{0.3118}$
 - $T_2 = 524.99 \text{ K}$
 - Now put the value of T_2 in eq (1)
 - $\Delta U = m \left(\frac{R}{\gamma-1}\right) (T_2 - T_1)$
 - $\Delta U = 1.4805 \left(\frac{287}{1.4-1}\right) (524.99 - 353)$
 - **$\Delta U = 182.698 \text{ kJ}$**
- Heat Transfer (Q):
 - $Q = \left(\frac{p_1 V_1 - p_2 V_2}{n-1}\right) \left(\frac{\gamma-n}{\gamma-1}\right)$
 - $Q = \left(\frac{(1.5 \times 10^5 \times 1) - (8 \times 10^5 \times 0.28)}{0.3118}\right) \left(\frac{1.4-1.3118}{1.4-1}\right)$
 - **$Q = -52.332 \text{ kJ}$ (Negative sign shows that heat is rejected by the air during the process)**

10. Explain steam formation with T-H diagram.

- Consider a cylinder fitted with a piston which can move freely upwards and downwards in it. Consider 1 kg of water at 0 deg C under the piston. As shown in below fig (4.1)(i).



- A weight W is placed over the piston so that it exerts constant pressure p on the water.
- This condition of water at 0 deg C is represented by the point A on the temperature – enthalpy graph as shown in below T-H diagram



- Now if the heat is supplied to water, a rise in temperature will be noticed and this rise will continue till boiling point is reached.

- When the boiling point of water is reached, there will be a slight increase in the water as shown in fig.
- The saturation temperature is defined as the temperature at which the water begins to boil at the stated pressure. This condition of water at temperature T_s is represented by the point B on the graph.
- The amount of heat required to raise the temperature of 1 kg of water from 0 deg C to the saturation temperature T_s deg C at a given pressure is known as the sensible heat and denoted by h_f . This heat is also called enthalpy of the liquid.
- Now, if supply of heat to water is continued it will eliminate the evaporation of water while the temperature remains at the saturation temperature T_s because the water will be saturated with heat and any further addition of heat changes only the phase from the liquid phase to the gaseous phase.
- This evaporation will be continued at the same saturation temperature T_s until the whole of the water is completely converted into steam as shown in fig. 4.1 (iv). This point is represented by the point C on the T- H diagram.
- This constant pressure and constant temperature heat addition is represented by the horizontal line BC on the T-H diagram.
- The heat being supplied does not show any rise of temperature but changes water into vapour state (steam) and is called latent heat or hidden heat or enthalpy of evaporation.
- If the steam is in contact with water, it is called wet steam (fig. 4.1 (iii)).
- Again, if supply of heat to the saturated steam is continued at constant pressure there will be increase in temperature and volume of steam. The temperature of the steam above the saturation temperature at a given pressure is called superheated temperature.
- During this process of heating, the dry steam will be heated from its dry state, and the process of heating is called superheating. The steam when superheated is called superheated steam.
- This superheating is represented by the inclined line CD on the T-H diagram.
- The amount of heat required to raise the temperature of dry steam from its saturation temperature to any required higher temperature at the given constant pressure is called amount of superheat or heat of superheat.
- The difference between the superheated temperature and the saturation temperature is known as degree of superheat.

11. Define the following terms: (a) Wet steam (b) Degree of superheat (c) Saturation temperature (d) Enthalpy of evaporation (f) Dryness Fraction (h) Wetness fraction (i) Latent heat

- **Wet steam:**
 - A wet steam is a two-phase mixture of entrained water molecules and steam in thermal equilibrium at the saturation temperature corresponding to a given pressure.
- **Degree of superheat:**
 - It is defined as the difference between the temperature of superheated steam and dry saturated steam at the given pressure. **Degree of superheat = $T_{sup} - T_{sat}$**

- **Saturation temperature:**

- The saturation temperature is defined as the temperature at which the water begins to boil at the stated pressure.

- **Enthalpy of evaporation:**

- The heat being supplied does not show any rise of temperature but changes water into vapour state (steam) and is called latent heat or hidden heat or enthalpy of evaporation.

- **Dryness Fraction:**

- It is the ratio of the actual mass of dry steam present in a known quantity of wet steam to the total mass of the wet steam.
- $x = \frac{m_s}{m_s + m_w}$
- Where, m_s = mass of dry steam present in the given quantity of wet steam.

m_w = mass of suspended water molecules present in the given quantity of wet steam.

- **Wetness fraction:**

- It is the ratio of the mass of water particles present in a known quantity of wet steam to the total mass of the wet steam.
- Wetness fraction = $\frac{m_w}{m_s + m_w} = 1 - x$

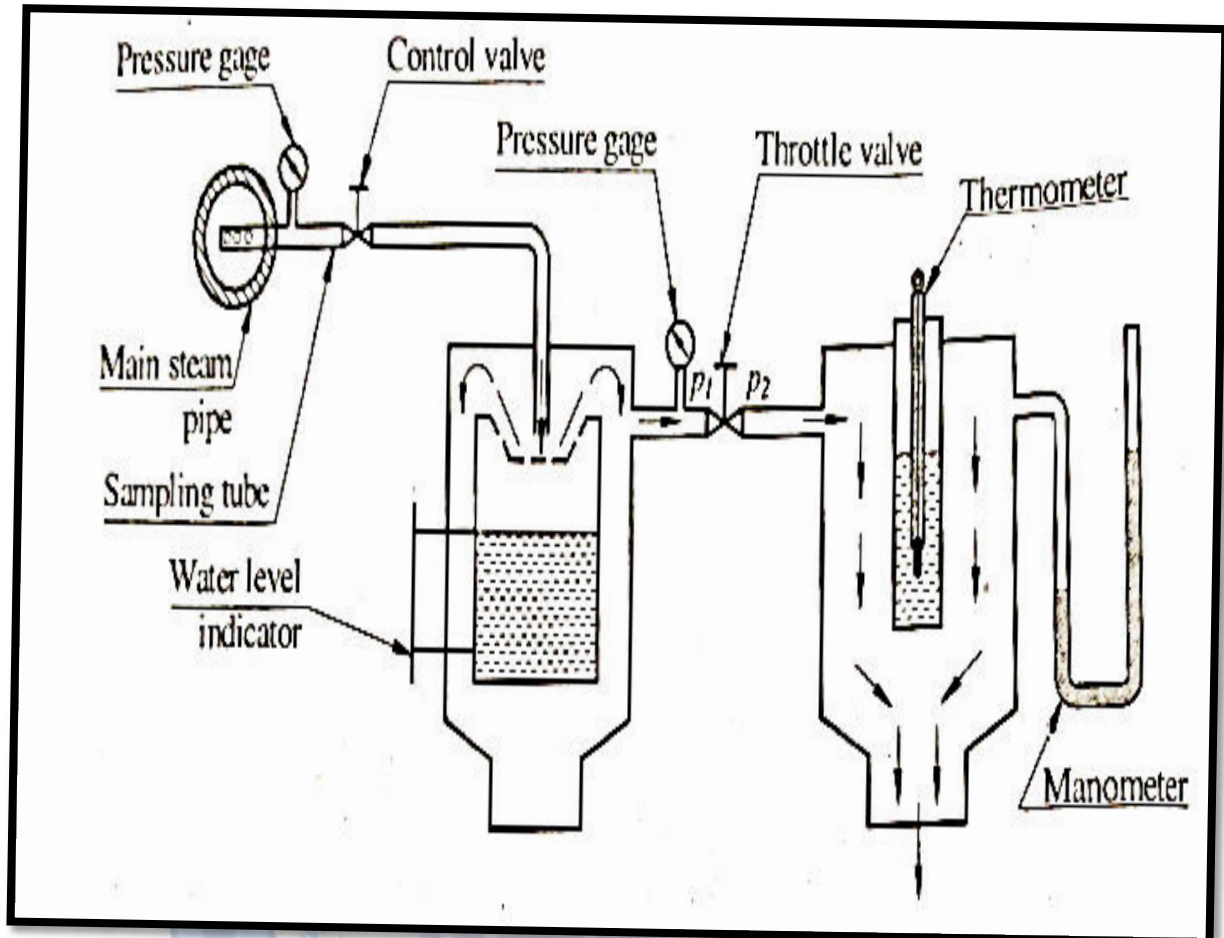
- **Latent heat:**

- The heat being supplied does not show any rise of temperature but changes water into vapour state (steam) is called latent heat or hidden heat or enthalpy of evaporation.

12. With a neat sketch explain structure and working of Combined separating and throttling calorimeter and derive the suitable equation for finding the dryness fraction of steam.

- It is already stated that the dryness fraction of the steam can be found by using throttling calorimeter only if the dryness fraction is greater than 0.9.
- And separating calorimeter gives only approximate result because of complete separation of moisture by mechanical means is not possible.
- So, a combination of separating and throttling calorimeter is used for accurate measurement of dryness fraction of steam.

In below figure combined separating and throttling calorimeter is represented.



- **Construction:**

- As we can see in this diagram throttling and separating calorimeter placed in series.

- **Working:**

- In this calorimeter, steam first passes through separating calorimeter.
- In separating calorimeter water particles are separated from wet steam and dryness fraction of steam increases.
- Now the steam enter into throttling calorimeter where steam becomes superheated after throttling.

- **Calculation of dryness fraction:**

- Let ,

x_1 = Dryness fraction of steam measured by separating calorimeter.

$$\therefore x_1 = \frac{m_s}{m_w + m_s}$$

Where,

m_s = mass of steam passing through separating calorimeter into throttling calorimeter.

m_w = mass of water collected in separating calorimeter

- Let ,

x_2 = Dryness fraction of wet steam which enter into throttling calorimeter.

$$\therefore x_2 = \frac{h_{g2} + C_{ps} * (T_{sup} - T_{sat}) - h_{f1}}{h_{fg1}}$$

- If x be the initial dryness fraction of steam then.

$$\therefore \text{Water droplet in the sample} = (1-x) (m_w + m_s) \text{ kg.}$$

$$\therefore \text{Water droplet removed by separating calorimeter} = (1-x_1) (m_w + m_s) \text{ kg}$$

$$\therefore \text{Water droplet passed through throttling calorimeter} = (1-x_2) m_s \text{ kg}$$

- water droplet in the sample =

Water droplet removed by separating calorimeter +

Water droplet passed through throttling calorimeter

$$\therefore (1-x) (m_w + m_s) = (1-x_1) (m_w + m_s) + (1-x_2) m_s$$

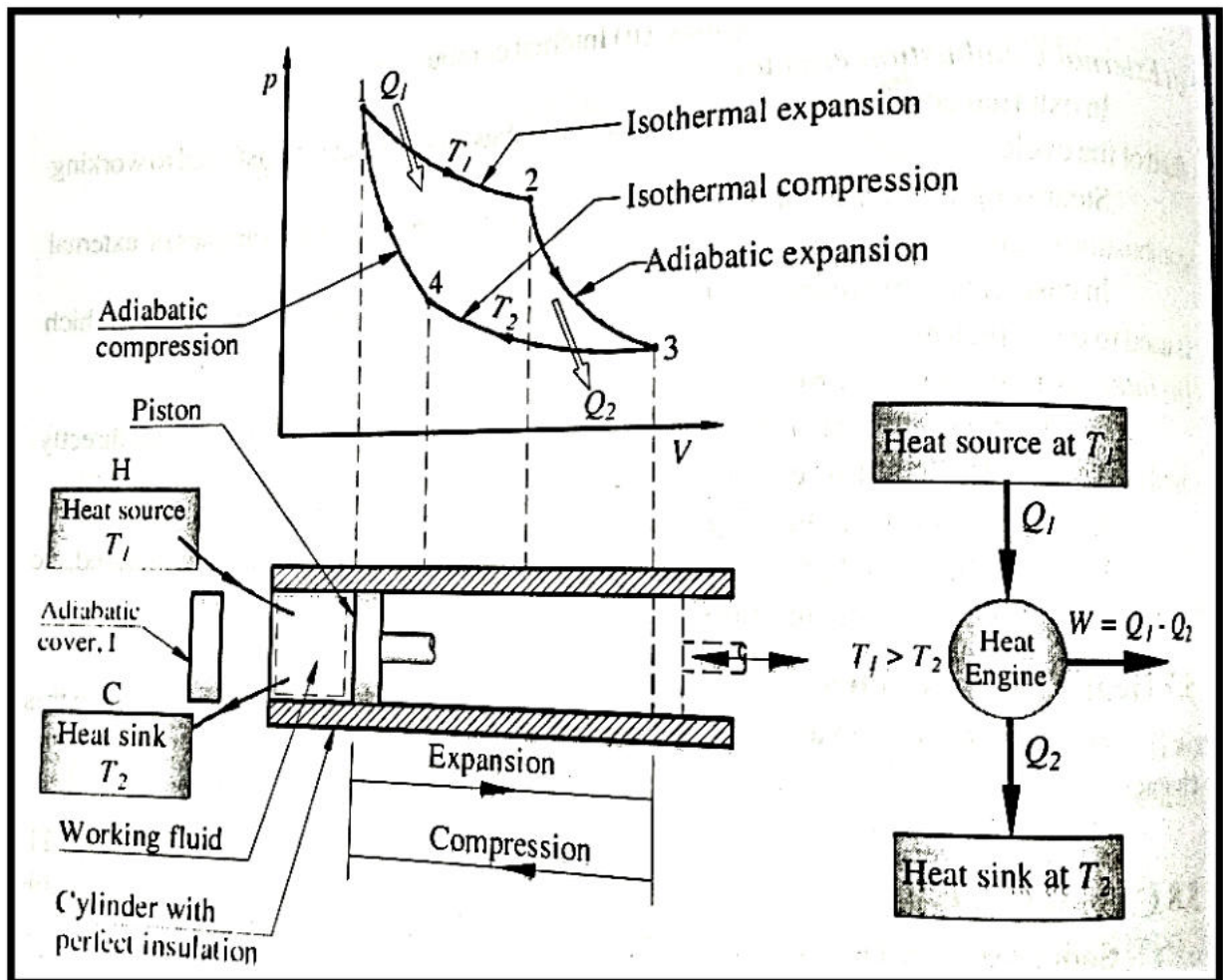
$$\therefore (1-x) = (1-x_1) + \frac{(1-x_2) m_s}{m_w + m_s}$$

$$\therefore 1-x = 1 - x_1 + x_1 + x_1 x_2$$

$$\therefore x = x_1 x_2$$

13. Explain Carnot cycle and derive expression for the efficiency of the Carnot cycle.

- According to Carnot theorem; **“No cycle can be more efficient than a reversible cycle operating between the same temperature limits.”**
- Carnot cycle is useful to compare the efficiency of any cycle under consideration with the efficiency of any cycle operating between the same two temperatures.
- A Carnot cycle is a hypothetical cycle consisting four different processes: two reversible isothermal processes and two reversible adiabatic (isentropic) processes.



• CARNOT CYCLE PROCESSES

- This cycle consists of four processes in the following order –
 1. Isothermal expansion (Process 1-2)
 2. Reversible Adiabatic (Isentropic) expansion (Process 2-3)
 3. Isothermal compression (Process 3-4)
 4. Reversible Adiabatic (Isentropic) compression (Process 4-1)
- **Isothermal Expansion Process (1-2)**
 - The source of heat is supplied to the end of the cylinder and isothermal reversible expansion occurs at T_1 . During this process Q_1 heat is supplied to the system.
- **Reversible Adiabatic (Isentropic) Expansion Process (2-3)**
 - The cylinder is perfectly insulated and the adiabatic cover (I) is brought in contact with the cylinder head. During this process, the system is allowed to expand adiabatically to final temperature T_2 .
- **Isothermal Compression Process (3-4)**
 - During this process, the adiabatic cover is removed and heat sink is applied to the end of the cylinder. The heat Q_2 is transferred reversibly and isothermally at temperature T_2 from the cylinder to the heat sink.

- **Reversible Adiabatic (Isentropic) Compression Process (4-1)**

- Adiabatic cover (I) is again brought in contact with the cylinder head. This completes the cycle and system is returned to its initial state 1. During this process, the temperature of the system is raised from T_2 to T_1 .

- **Efficiency Of Carnot Cycle**

- Consider 1 kg mass of the working substance.
- Heat supplied during isothermal expansion process (1-2) is given as

$$Q_1 = p_1 * V_1 * \ln \left(\frac{V_2}{V_1} \right)$$

$$\text{Thus } Q_1 = R * T_1 * \ln \left(\frac{V_2}{V_1} \right) \quad \text{----- (1)}$$

- Heat rejected during isothermal compression process (3-4) is given as

$$Q_2 = p_3 * V_3 * \ln \left(\frac{V_4}{V_3} \right)$$

$$\text{Thus } Q_2 = R * T_2 * \ln \left(\frac{V_4}{V_3} \right) \quad \text{----- (2)}$$

- During adiabatic expansion (2-3) and adiabatic compression (4-1), the heat transfer from or to the system is zero.

- Let,

$$r = \text{ratio of expansion for process (1-2)} = V_2 / V_1$$

$$\text{Also, } r = \text{ratio of compression for process (3-4)} = V_4 / V_3$$

- Now substituting the value of r in equations (1) and (2), we have

Thus,

$$\therefore Q_1 = R * T_1 * \ln(r) \quad \text{and} \quad Q_2 = R * T_2 * \ln(r)$$

- Now net work done during the cycle, $W = \text{Heat supplied } (Q_1) - \text{Heat rejected } (Q_2)$

Thus,

$$W = R * T_1 * \ln(r) - R * T_2 * \ln(r) = R * \ln(r) * (T_1 - T_2)$$

- Now the Thermal efficiency of Carnot cycle,

$$\eta = \frac{\text{Net Work done } (W)}{\text{Heat supplied } (Q_1)} = \frac{R * \ln(r) * (T_1 - T_2)}{R * T_1 * \ln(r)} = \frac{(T_1 - T_2)}{T_1}$$

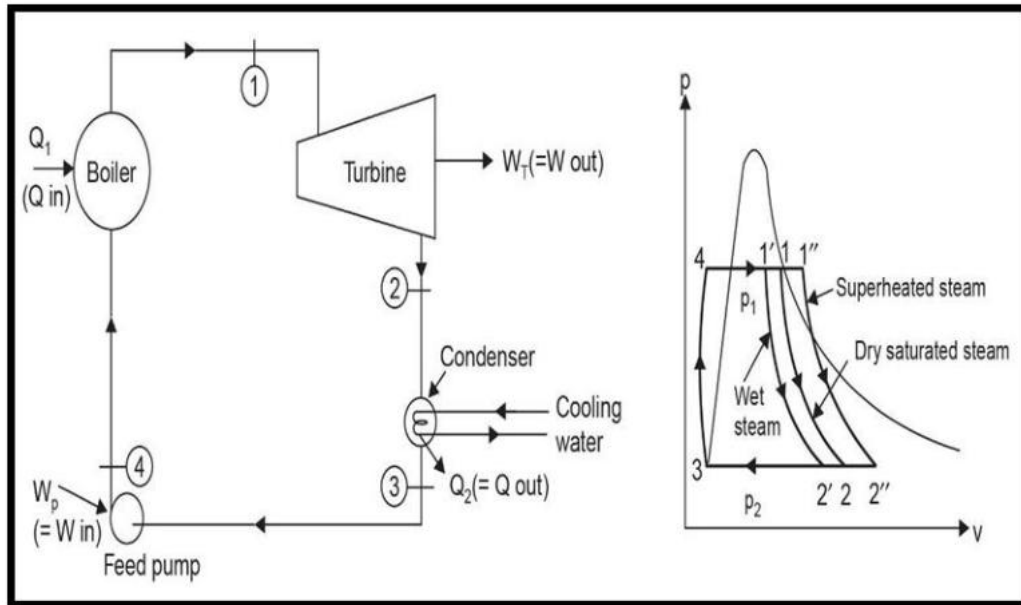
Thus,

$$\eta = 1 - \left(\frac{T_2}{T_1} \right)$$

Where T_1 is the maximum temperature of the cycle (Source Temperature) and T_2 is the minimum temperature of the cycle (Sink Temperature).

14. Explain working of Rankine cycle with P-V diagram. Derive the formula for efficiency of Rankine cycle.

- The Rankine cycle is the ideal cycle for steam power plants. In steam power plants, the heat energy of the fuel is converted into mechanical energy or power.
- The ideal Rankine cycle is shown schematically and on P-V diagram in Fig.



• RANKINE CYCLE PROCESSES

• Process 4 – 1: Constant pressure heat addition in the boiler.

- The water is heated at constant pressure p_1 in the boiler until the saturation temperature is reached. Saturated water is converted into saturated steam at constant pressure. During 1-1' process steam is superheated in super heater.

∴ Heat supplied to the Boiler is given as $Q_S = (h_1 - h_4)$

• Process 1 – 2: Isentropic expansion in the Steam turbine.

- High pressure and high temperature superheated, dry saturated or wet steam generated in the boiler at p_1 and T_1 is supplied to the steam turbine. This steam expands isentropically into steam turbine up to the condenser pressure. Steam turbine develops mechanical work, W_T due to expansion of steam.

∴ Work developed by the Turbine is given as $W_T = (h_1 - h_2)$

• Process 2 – 3: Constant pressure heat rejection in the condenser.

- The exhaust steam from turbine enters into condenser, where it is condensed at constant pressure by circulating cooling water in the tubes. The heat rejected by exhaust steam is Q_R .

∴ Heat rejected in the Condenser is given as $Q_R = (h_2 - h_3)$

• Process 3 – 4: Isentropic compression in the Feed pump.

- The condensed water coming from condenser is pumped to boiler at boiler pressure with the help of feed pump. For the operation of the feed pump, work is supplied to the pump, W_P .

∴ Work supplied to the Feed pump is given as $W_P = (h_4 - h_3)$

- **Efficiency Of Rankine Cycle**

- Consider 1 kg mass of the working substance.
- Net work done during the cycle, (W_{net}) = Turbine work (W_T) – Pump work (W_P)
- Usually the pump work is very small as compared to turbine work. Hence the pump work is neglected.
- Thus,

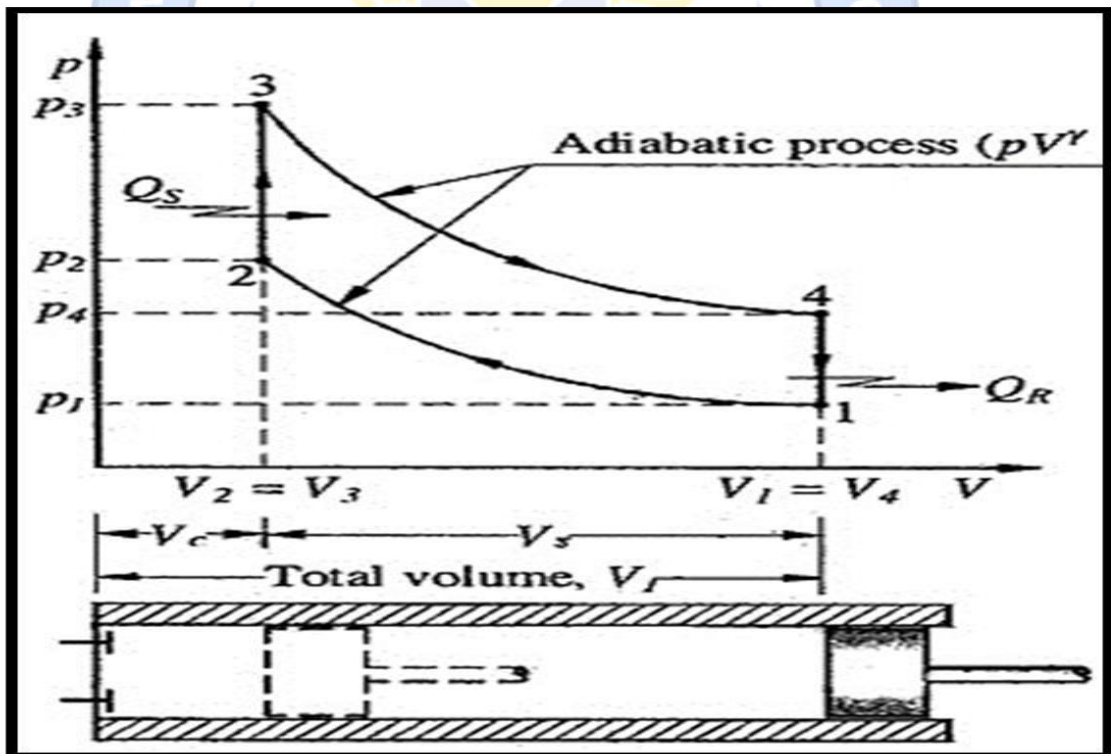
$$(W_{net}) = \text{Turbine work } (W_T) = (h_1 - h_2)$$

- Now the Efficiency of Rankine cycle,

$$\eta = \frac{\text{Net Work done } (W_{net})}{\text{Heat supplied to Boiler } (Q_s)} = \frac{h_1 - h_2}{h_1 - h_4}$$

15. Show that the efficiency of Otto cycle is a function of Compression Ratio only.

- The Otto cycle consists of two constant volume and two adiabatic processes as shown in figure.



- **EFFICIENCY OF OTTO CYCLE**

- The amount of heat supplied during constant volume process (2-3) is given as

$$\therefore Q_s = m * C_v * (T_3 - T_2)$$

- Similarly, the amount of heat rejected during constant volume process (4-1) is given as

$$\therefore Q_R = m * C_v * (T_4 - T_1)$$

- Net work done during the cycle,

$$\therefore (W_{\text{net}}) = \text{Heat supplied } (Q_S) - \text{Heat rejected } (Q_R)$$

- Thus,

$$(W_{\text{net}}) = m * C_v * (T_3 - T_2) - m * C_v * (T_4 - T_1)$$

- Now the air standard efficiency of Otto cycle,

$$\therefore \eta = \frac{\text{Net Work done } (W_{\text{net}})}{\text{Heat supplied } (Q_S)} = \frac{m * C_v * (T_3 - T_2) - m * C_v * (T_4 - T_1)}{m * C_v * (T_3 - T_2)}$$

$$\therefore \eta = 1 - \frac{T_4 - T_1}{T_3 - T_2} \quad \text{----- (1)}$$

- Compression ratio is defined as $r = \frac{V_c + V_s}{V_c}$

But from the p-V diagram, $V_c + V_s = V_1$ and $V_c = V_2$

Thus, the compression ratio, $r = \left(\frac{V_1}{V_2} \right)$

- Now expansion ratio = $\left(\frac{V_4}{V_3} \right)$

Since $V_4 = V_1$ and $V_3 = V_2$

Thus, Expansion ratio = $r = \left(\frac{V_1}{V_2} \right)$

- Now for Adiabatic compression process (1-2) we get,

$$\therefore (T_2 / T_1) = (V_1 / V_2)^{(\gamma-1)} = (r)^{(\gamma-1)}$$

$$\therefore T_1 = T_2 / (r)^{(\gamma-1)}$$

- Now for Adiabatic expansion process (3-4) we get,

$$\therefore (T_3 / T_4) = (V_4 / V_3)^{(\gamma-1)} = (r)^{(\gamma-1)}$$

$$\therefore T_4 = T_3 / (r)^{(\gamma-1)}$$

- Substituting these values of T_1 and T_4 in equation (1)

Hence,

$$\eta = 1 - \frac{T_3 / (r)^{(\gamma-1)} - T_2 / (r)^{(\gamma-1)}}{T_3 - T_2}$$

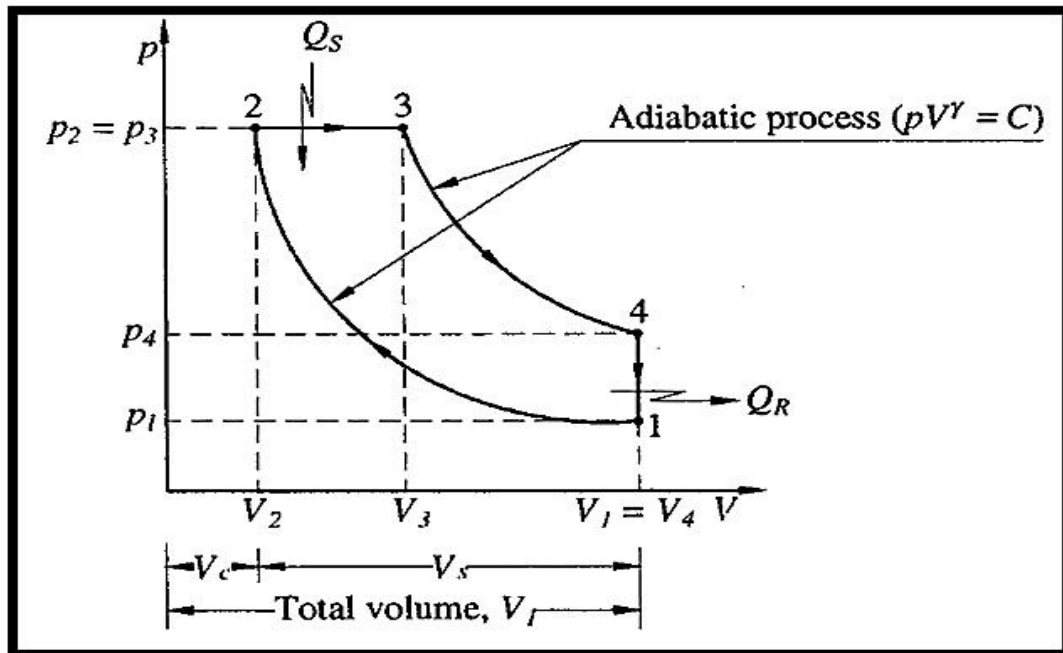
- Therefore, the Air standard efficiency of Otto cycle,

$$\eta = 1 - \frac{1}{(r)^{(\gamma-1)}} \quad \text{----- (2)}$$

- From above equation (2), it is proved that the air standard efficiency of Otto cycle depends only on the compression ratio (r).

16. Derive an expression of air standard efficiency of Diesel cycle.

- The Diesel cycle consists of two reversible adiabatic process, a constant pressure heat addition process and a constant volume heat rejection process as shown in figure.



• EFFECIENCY OF DIESEL CYCLE

- The amount of heat supplied during constant pressure process (2-3) is given as
 $\therefore Q_S = m * C_p * (T_3 - T_2)$
- Similarly, the amount of heat rejected during constant volume process (4-1) is given as
 $\therefore Q_R = m * C_v * (T_4 - T_1)$

- Net work done during the cycle, $(W_{net}) = \text{Heat supplied } (Q_S) - \text{Heat rejected } (Q_R)$
- Thus,

$$(W_{net}) = m * C_p * (T_3 - T_2) - m * C_v * (T_4 - T_1)$$

- Now the air standard efficiency of Diesel cycle,

$$\eta = \frac{\text{Net Work done } (W_{net})}{\text{Heat supplied } (Q_S)} = \frac{m * C_p * (T_3 - T_2) - m * C_v * (T_4 - T_1)}{m * C_p * (T_3 - T_2)}$$

Thus,

$$\therefore \eta = 1 - \left(\frac{1}{\gamma}\right) * \left(\frac{T_4 - T_1}{T_3 - T_2}\right) \quad \text{----- (1)}$$

- Compression ratio is defined as $r = \frac{V_c + V_s}{V_c}$

But from the p-V diagram, $V_c + V_s = V_1$ and $V_c = V_2$

Thus, the compression ratio, $r = \left(\frac{V_1}{V_2}\right)$

- The cut-off ratio, $\rho = \left(\frac{V_3}{V_2} \right)$
- Now expansion ratio $= \left(\frac{V_4}{V_3} \right) = \left(\frac{V_4/V_2}{V_3/V_2} \right)$

Since, $V_4 = V_1$ from the p-V diagram

$$\text{Thus Expansion ratio} = \left(\frac{V_1/V_2}{V_3/V_2} \right) = \left(\frac{r}{\rho} \right)$$

- Now for Adiabatic compression process (1-2) we get

$$\therefore (T_2 / T_1) = (V_1 / V_2)^{(\gamma-1)} = (r)^{(\gamma-1)}$$

$$\therefore T_2 = T_1 * (r)^{(\gamma-1)}$$

- Now for constant pressure process, (2-3) we get by applying Charles' Law,

$$\therefore (V_2 / T_2) = (V_3 / T_3)$$

$$\therefore T_3 = T_2 * (V_3 / V_2)$$

Now substituting the values of T_2 and (V_3 / V_2) in above equation, we get

$$\therefore T_3 = T_1 * (r)^{(\gamma-1)} * \rho$$

- Now for Adiabatic expansion process (3-4) we get

$$\therefore (T_4 / T_3) = (V_3 / V_4)^{(\gamma-1)} = (\rho/r)^{(\gamma-1)}$$

Substituting the value of T_3 in the above equation and doing mathematical calculations we get,

$$\therefore T_4 = T_1 * (\rho)^\gamma$$

- By substituting the values of T_2, T_3, T_4 in equation (1) and doing the mathematical calculations, we finally get the Air standard efficiency of Diesel cycle,

$$\therefore \eta = 1 - \left[\frac{1}{(r)^{(\gamma-1)}} \right] * \left[\frac{(\rho)^\gamma - 1}{\gamma(\rho-1)} \right] \quad \text{----- (2)}$$

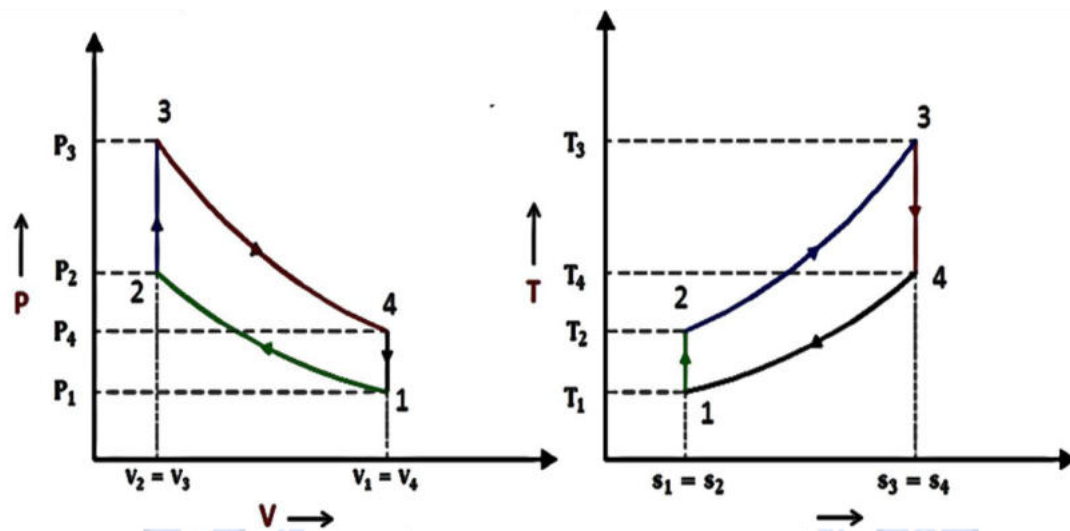
- It is clear from the above equation that the air standard efficiency of Diesel cycle depends upon the compression ratio, the cut-off ratio and the ratio of the specific heats.

17. In an ideal Otto cycle, the air at the beginning of isentropic compression is at 1 bar and 15 deg C. The ratio of compression is 8. If the heat added during the constant volume process is 1000 kJ/kg, determine (a) the maximum temperature in the cycle, (b) the air standard efficiency (c) work done per kg of air.

• **Given Data:**

- Pressure of air at compression beginning, $p_1 = 1 \text{ bar} = 10^5 \text{ Pa}$
- Temperature of air at compression beginning, $T_1 = 15 \text{ deg C} = 288 \text{ K}$

- Compression ratio, $r = 8$ (Ideal Otto cycle)
- Amount of heat supplied, $q_s = 1000 \text{ kJ/kg} = 1000 \times 10^3 \text{ J/kg}$
- **To Find:**
 - Maximum temperature; T_3
 - Air standard efficiency; η_{otto}
 - Work done per kg air; w



- **Calculations:**
 - Air standard efficiency for Otto cycle (η_{otto}):

- $\eta_{otto} = 1 - \frac{1}{(r)^{\gamma-1}}$
- $\eta_{otto} = 1 - \frac{1}{(8)^{1.4-1}}$
- $\eta_{otto} = 56.47 \%$
- **$\eta_{otto} = 56\%$**

- Work done per kg of air (w):

- $w = \eta_{otto} \times q_s$
- $w = 0.5647 \times 1000 \times 10^3$
- **$w = 564.7 \text{ kJ/kg}$**

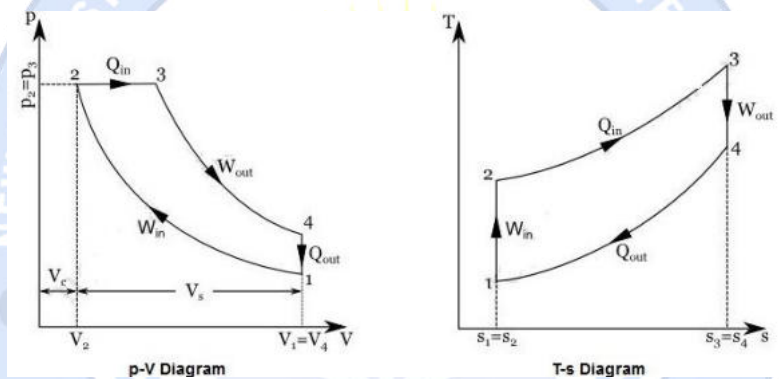
- Maximum temperature (T_3):

Now amount of heat supplied during constant volume process,
 $q_s = C_v(T_3 - T_2) \dots (1)$

- For T_2 ,
- $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

- $T_2 = T_1(r)^{\gamma-1}$
- $T_2 = 288(8)^{0.4}$
- $T_2 = 661.65 \text{ K}$
- Now put the value of T_2 in eq (1)
- $1000 \times 10^3 = 718 (T_3 - 661.65) \quad \left(\because C_v = 718 \frac{\text{J}}{\text{kg K}}, \text{ for air} \right)$
- **$T_3 = 2054.41 \text{ K}$**

18. An engine operating on the ideal diesel cycle has a maximum pressure of 45 bar and a maximum temperature of 1500 deg C. The pressure and temperature of air at the beginning of compression stroke are 1 bar and 27 deg C respectively. Find the air standard efficiency of the cycle. Assume $\gamma = 1.4$



• **Given Data:**

- Maximum pressure, $P_3 = 45 \text{ bar} = 45 \times 10^5 \text{ Pa}$
- Maximum Temperature, $T_3 = 1500 \text{ deg C} = 1773 \text{ K}$
- Pressure at compression beginning, $P_1 = 1 \text{ bar} = 1 \times 10^5 \text{ Pa}$
- Temperature at compression beginning, $T_1 = 27 \text{ deg C} = 300 \text{ K}$
- $\gamma = 1.4$

• **To Find:**

- Air standard efficiency of diesel cycle; η_{diesel}

• **Calculation:**

- Air standard efficiency (η_{diesel}):

- $\eta_{\text{diesel}} = 1 - \frac{1}{(r)^{\gamma-1}} \left(\frac{\rho^{\gamma}-1}{\gamma(\rho-1)} \right) \dots (a)$

- For r;

- $\frac{p_2}{p_1} = \left(\frac{V_1}{V_2} \right)^{\gamma}$

- $r = \left(\frac{p_2}{p_1}\right)^{\frac{1}{\gamma}} \quad \left(\because r = \frac{V_1}{V_2}\right)$

- $r = \left(\frac{45 \times 10^5}{1 \times 10^5}\right)^{\frac{1}{1.4}}$

- **$r = 15.166$**

- For ρ ;

- $\rho = \frac{V_3}{V_2} = \frac{T_3}{T_2} \dots (1)$

- For T_2 ;

- $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

- $\frac{T_2}{T_1} = (r)^{\gamma-1}$

- $T_2 = T_1(r)^{\gamma-1}$

- $T_2 = 300(15.166)^{0.4}$

- **$T_2 = 890.16 \text{ K}$**

- put value of T_2 in eq (1)

- $\rho = \frac{1773}{890.16}$

- **$\rho = 1.992$**

- Putting value of r and ρ in eq (a)

- $\eta_{\text{diesel}} = 1 - \frac{1}{(15.166)^{0.4}} \left(\frac{1.992^{1.4}-1}{1.4(1.992-1)} \right)$

- $\eta_{\text{diesel}} = 60.60 \%$

- **$\eta_{\text{diesel}} = 61 \%$**

19. What is boiler? Compare fire tube boiler with water tube boiler.

- Steam boiler is a closed vessel in which heat produced by the combustion of fuel is utilized to generate steam from water, at desired temperature and pressure.
- According to IBR (Indian Boiler Regulation) Act 2007, boiler is defined as **“Boiler is a closed pressure vessel with capacity exceeding 22.75 litres used for generating steam under pressure.”**

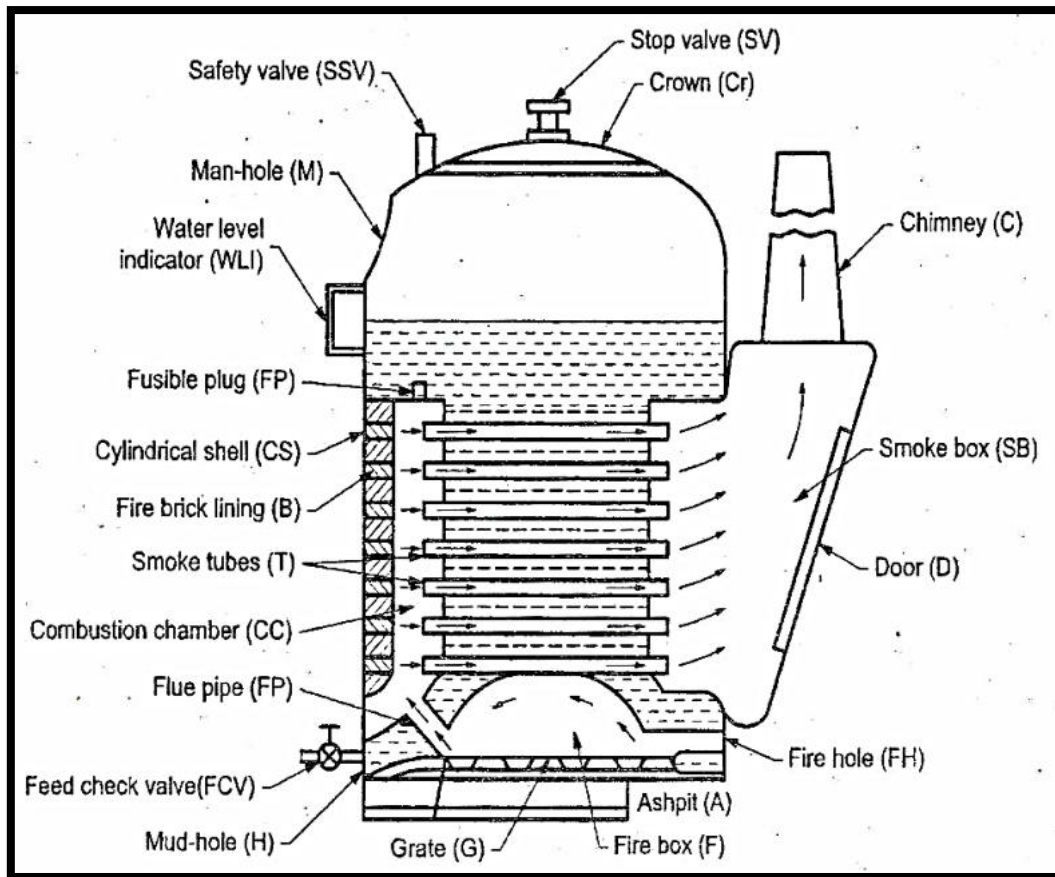
- Difference between fire tube boiler and water tube boiler mentioned below.

Parameters	Fire tube Boiler	Water tube Boiler
Position of water and hot gases	Hot gases inside the tubes and water outside the tubes	Water inside the tubes and hot gases outside the tubes
Operating pressure	Limited to 25 bar	More than 125 bar
Rate of steam generation	Lower	Higher
Suitability	Not suitable for large power plants	Suitable for large power plants
Floor space requirement	More	Less
Cost	Less	More
Requirement of skill	Required less skill for efficient and economic operation	Required more skill and careful attention for efficient and economic operation
Use	For producing process steam	For producing steam for power generation and process heating
Scale deposition and overheating	Since there are no water tubes, no problem of scale deposition and less problem of overheating and bursting	Water tubes are present, hence small deposition of scale will cause overheating and bursting of tubes

20. Discuss construction and working of Cochran boiler with sketch.

- Cochran boiler is one of the best type of vertical multi-tubular boiler. It is fire tube boiler and used for steam generation at lower rate.
- Cochran boiler shown in below figure.





- **Construction**

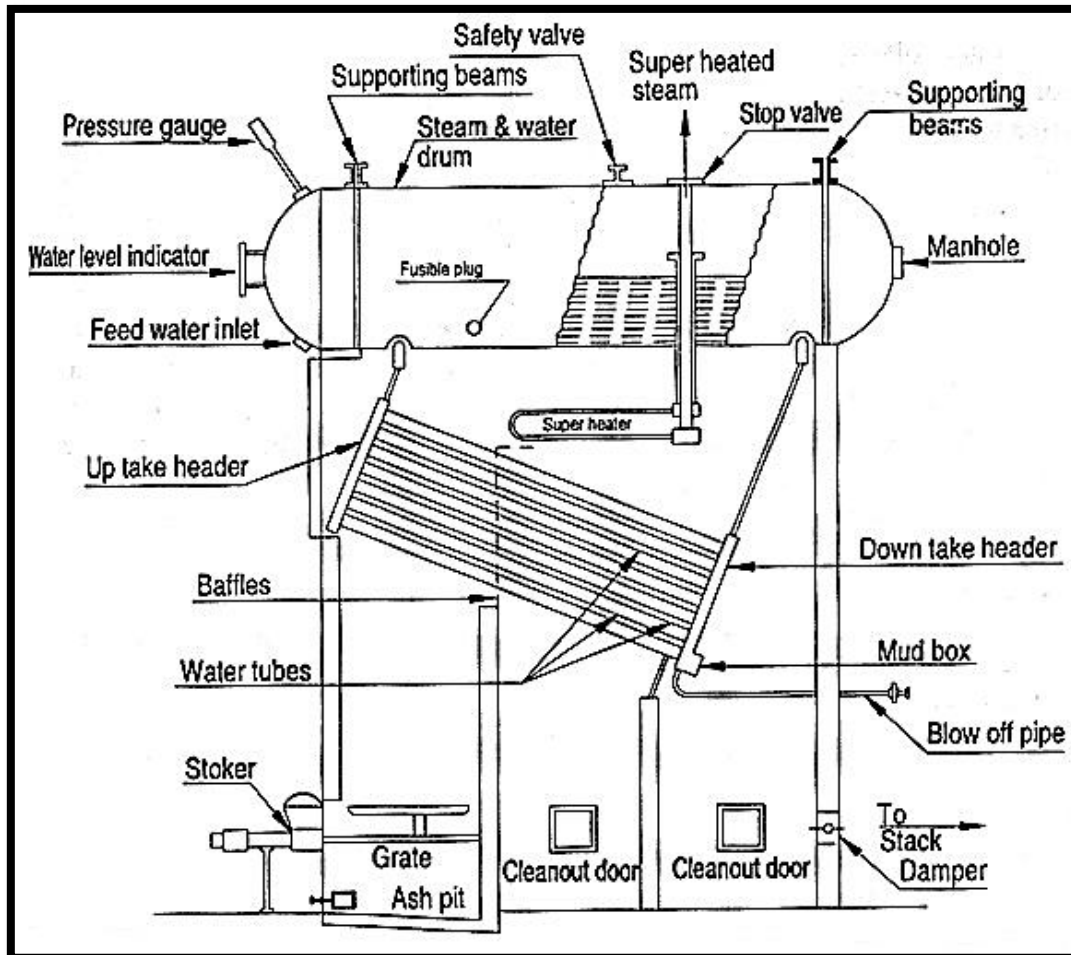
- The boiler consists of a cylindrical shell, hemispherical fire box, fire tubes and chimney. The top of the shell having hemispherical shaped crown as shown in figure.
- The hemispherical crown of boiler gives good strength to withstand pressure of steam inside the boiler. The hemispherical shape of furnace can withstand high heat and it is also useful to increase radiant heat transfer.
- The grate is placed at the bottom of furnace and ash pit is located below the grate.
- The furnace and the combustion chamber are connected by short flue pipe. The wall of the combustion chamber is lined with the fire bricks.

- **Working**

- The water is supplied to the boiler through feed check valve.
- The level is adjusted with the help of water level indicator. Coal is added through the fire-hole or door to the grate and burnt. The hot gases produced are collected in the fire box.
- These hot gases enter into horizontal fire tubes. Heat transfer takes place from flue gases passing inside the tubes to water surrounded the tubes.
- The flue gas coming from the fire tubes enter into smoke box. Finally, they discharged to atmosphere through a chimney.
- The ash formed is collected in ash pit.
- The steam is collected through anti priming pipe on the top of the shell and is removed from the boiler through the Steam stop valve.

21. Explain Babcock and Wilcox Boiler with neat sketch.

- It is a water tube boiler and used in stationary and marine engine.
- The efficiency of this boiler is much greater than that of the fire tube boiler.
- This boiler is exclusively used when pressure is above 10 bar and steam generating capacity is required higher than 7000 kg/hr.



• Construction

- Fig. shows a Babcock and Wilcox boiler. It consists of inclined water tubes, a steam and water drum, a mud box and super heater.
- The drum is connected with uptake and downtake header by short riser tubes. These headers are connected to series of inclined water tubes.
- The water tubes are inclined to horizontal about 15° or above to bring natural circulation of water.
- The hand hole is provided in header in front of each tube for cleaning and inspection of tubes.
- The baffle plates are provided in order to make the circulation of hot gases in sine wave form. A damper is fitted at back of the boiler to regulate the draught and chain grate stoker is used to feed the coal to furnace.
- Soot doors are provided to clean the outside of water tubes and to remove the soot. Soot doors are also helpful to access the interior of the boiler.

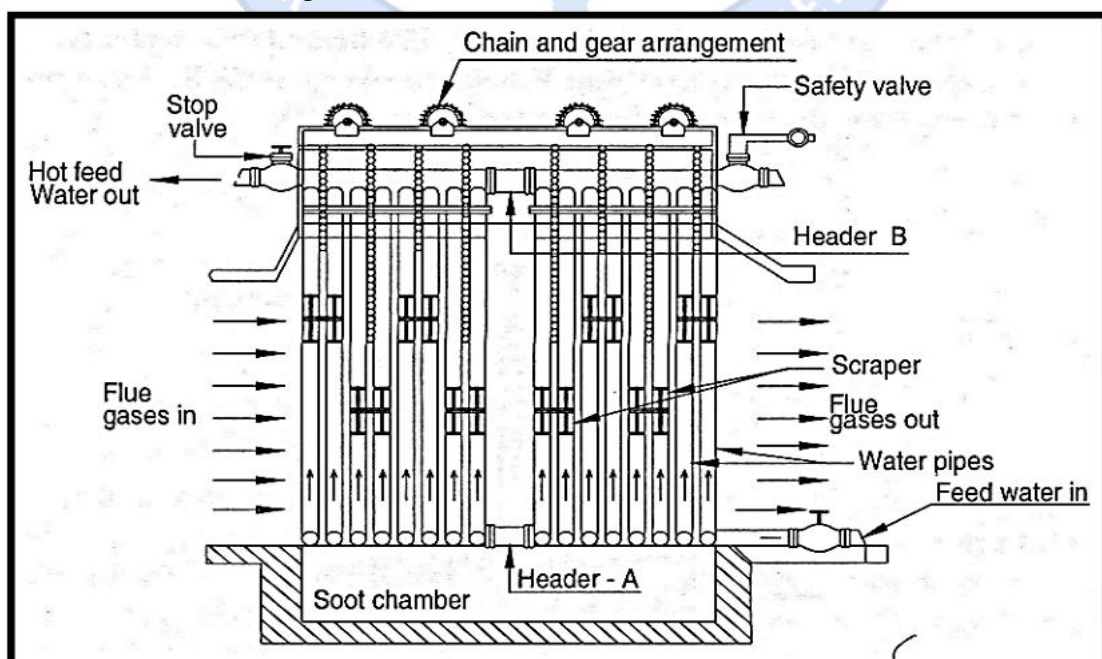
- **Working**

- The water fed into the boiler shell through the feed check valve. Due to gravity, water passes through the vertical tubes, headers and fills up the inclined tubes first.
- Then the water collects in the drum. Initially one half of drum is filled up with water.
- The coal is introduced to furnace grate by help of stoker. The coal is fired; hot gases produced are first forced to move upwards through passage between tubes.
- The baffle plates make flow of hot gases in sine wave, as move downwards and then move upwards over the water tubes. The damper controls the flow of air into the furnace.
- Water in the drum comes down through downtake header and enters the tubes. They are heated by hot gases coming from furnace.
- Due to heating of the water, density of water decreases. Low density water moves upward in water tubes. The water tubes just above the furnace are heated comparatively at a higher temperature than the rest of it.
- Therefore low density water is gradually converted into steam in their path and rises into drum through uptake header.
- Thus a continuous circulation of water from drum to water tubes and water tubes to drum is maintained due to density difference of water and gravity, without any pump.
- The steam then enters to the antipriming pipe and flows in the superheater tubes where it is further heated and finally the superheated steam is taken out through the main steam stop valve and supplied to the engine when needed.

22. Explain Economizer and Fusible plug with neat sketch.

- **Economizer**

- **Function** – An economizer is a device in which the waste heat of the flue gases is utilized for heating the feed water.

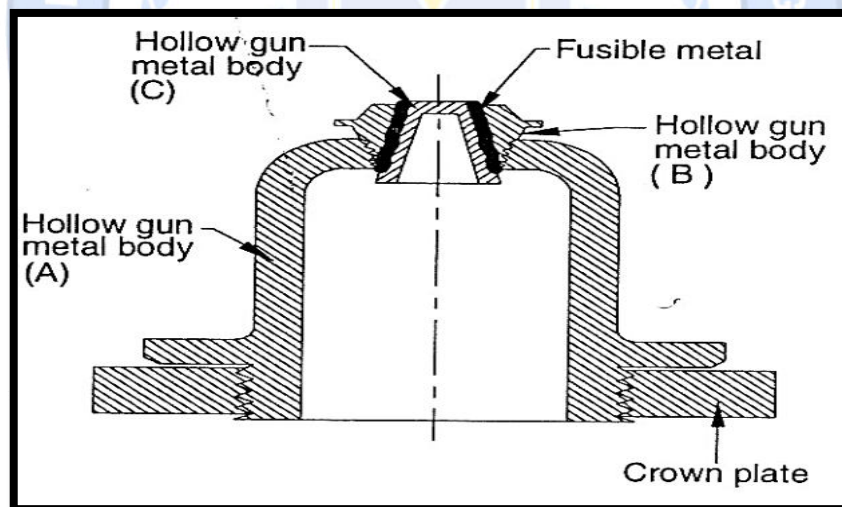


- **Construction and Working –**

- Fig. shows an independent type vertical tube economiser. It is employed for boilers of medium pressure range upto about 25 bar.
- It consists of a large number of vertical cast iron pipes which are connected through two horizontal pipes, one at the top and other at the bottom.
- A is the bottom pipe through which the feed water is pumped into the economizer. The water comes into the top pipe B from the bottom pipe and finally flows into the boiler.
- The flue gases flows around the pipes in the direction opposite to the flow of water. Consequently, heat transfer to the surface of the pipes takes place and water is thereby heated.
- A blow off cock is provided at the back end of vertical pipes to remove sediments deposited in the bottom boxes.
- The soot of flue gases deposited on the pipes reduces the efficiency of economizer. To prevent the soot deposit, the scrapers move up and down to keep the external surface of pipe clean.

- **Fusible Plug**

- **Function** – The main function of the fusible plug is to extinguish fire when water level in the boiler falls below an unsafe level.



- **Construction –**

- The construction of the fusible plug is shown in Figure which consists of three plugs.
- The hollow plug A having hexagonal flanges is screwed to the fire box crown plate.
- The plug B gunmetal plug is screwed to the body A.
- The third plug C is made up of copper is locked with metal like tin or lead which has a low melting point.

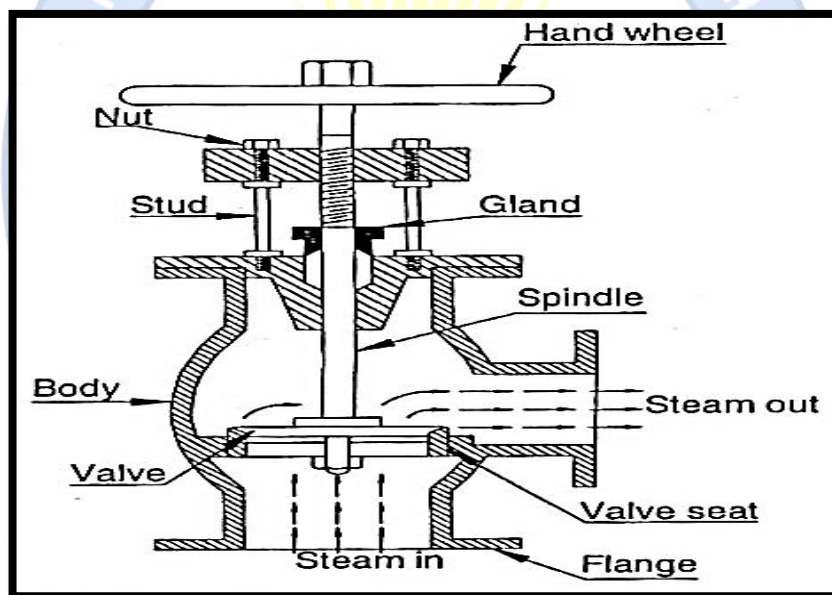
- **Working –**

- In normal working condition, water covers the fusible plug. Hence the plug remains cool.
- In case the water level falls below the danger levels, the fusible plug gets exposed to steam. This overheats the plug and fusible metal having low melting point melts quickly.
- Due to this the plug 'C' falls down.
- As a result, a path through hollow gun metal body 'A' opens for steam and hence the steam rushes into the fire box and extinguish the fire. Thus it stops the further steam generation.

23. Explain Steam stop valve and Air preheater with neat sketch.

- **Steam Stop Valve**

- **Function –** To control the flow of steam from the boiler to the main steam pipe and to shut off the steam completely when required.

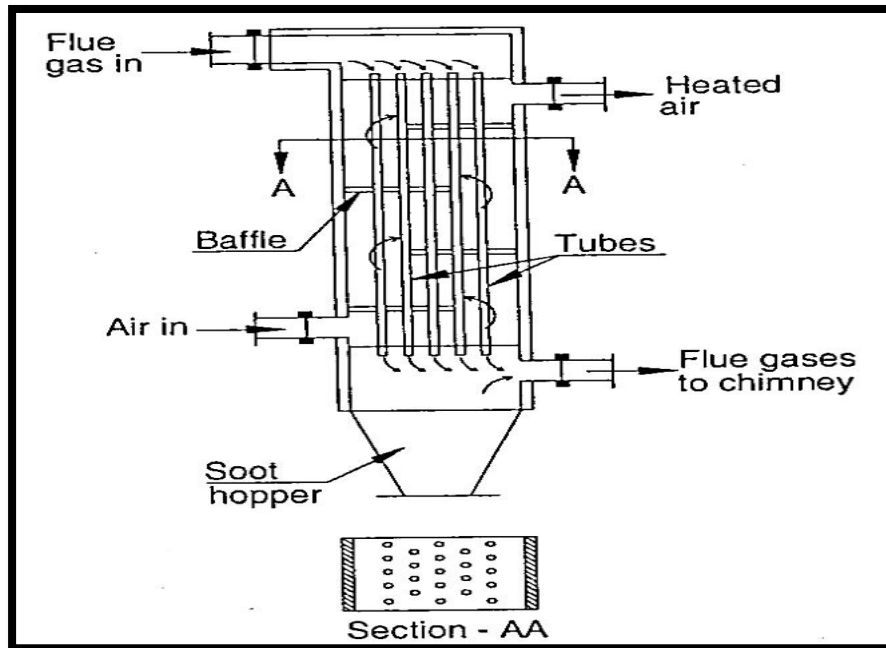


- **Construction and Working –**

- The body of the stop valve is made of cast iron or cast steel.
- The valve, valve seat and the nut through which the valve spindle works, are made of brass or gun metal.
- The spindle passes through a gland and stuffing box. The spindle is rotated by means of a hand wheel.
- The rotation of the spindle causes the valve to move up and down.
- When the valve sits over the valve seat, the passage of steam is completely closed.
- The passage may be partially or fully opened for the flow of steam by moving the valve up, rotating the hand wheel.

- **Air Preheater**

- **Function** – Air preheater increase the temperature of air before it supply to the furnace using heat from flue gases passing through the chimney.



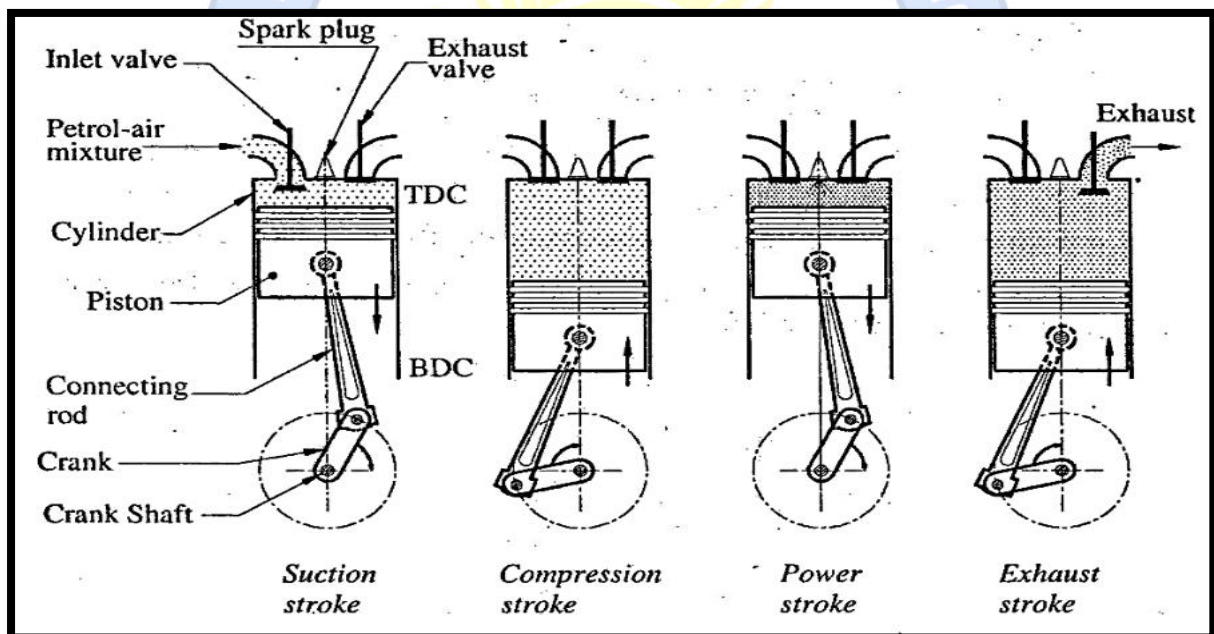
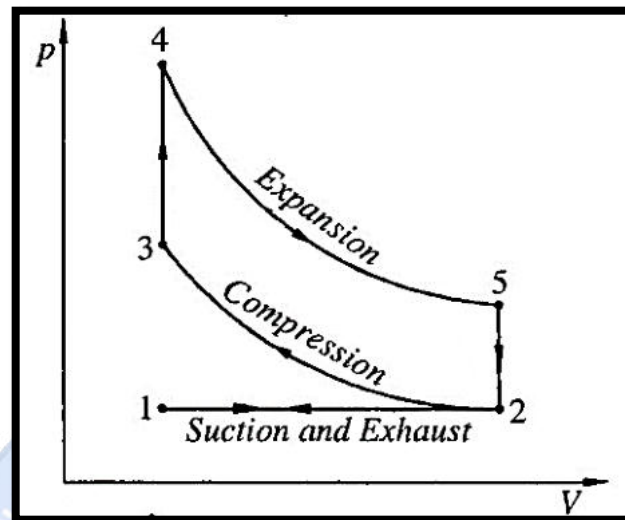
- **Construction and Working –**

- Air preheater is installed between economizer and the chimney.
- It consists of large numbers of tubes which arranged in path of flue gases as shown in Figure.
- Hot flue gases enters into tubes from top of shell and leave from the bottom to the chimney.
- The inlet air at room temperature is admitted into shell at lower end with the help of the fan. The air passes upward around the tubes in the opposite direction to the flow of hot flue gases. During these flow, the air absorbs the heat from the flue gases and is thus preheated.
- The soot hopper provided at the bottom is used to collect soot during cleaning operation of the tubes

24. Discuss the construction and working of four stroke Petrol engines.

- This engine works on Otto cycle and uses petrol (or gas) as a fuel.
- In this engine, spark is produced to ignite the charge.
- This type of engines work on constant volume combustion cycle as combustion takes place nearly at constant volume with increase of pressure.
- In a petrol engine, the petrol is evaporated and also it is mixed with correct proportion of air by the device which is known as carburettor.
- The whole cycle is completed in four strokes, namely
 1. Suction stroke

2. Compression stroke
 3. Working or Power or Expansion stroke
 4. Exhaust stroke
- All these strokes are represented as follows on PV diagram and the operating diagrams.



1. SUCTION STROKE –

- During this stroke, inlet valve remains open and exhaust valve is closed, the pressure in the cylinder will be atmospheric.
- The piston moves from TDC to BDC. The Crank rotation is from 0 to 180 deg.
- So the volume in the cylinder increases, while simultaneously the pressure decreases.
- This creates a pressure difference between the atmosphere and inside of the cylinder.

- Due to this pressure difference, the petrol and air mixture will enter into the cylinder through carburettor.
- This stroke is represented by the horizontal line 1-2 on the p-v diagram.
- At the end of this stroke piston reaches at BDC, the cylinder will be filled completely with petrol and air mixture called charge and inlet valve is closed.

2. COMPRESSION STROKE -

- As shown in fig., during this stroke both the inlet valve and exhaust valve remains closed. Piston moves from BDC to TDC. The Crank rotation is from 180 to 360 deg.
- As this stroke is being performed, the petrol and air mixture contained in the cylinder will be compressed, so pressure and temperature of mixture increases.
- The process of compression is shown in fig. by the curve 2-3.
- Near the end of this stroke, the petrol and air mixture is ignited by electric spark given out by the spark plug.
- The combustion of the petrol releases the hot gases which will increase the pressure at constant volume.
- This constant volume combustion process is represented by the vertical line 3-4 on the p-V diagram.

3. EXPANSION OR WORKING OR POWER STROKE -

- During this stroke, both the inlet valve and exhaust valve remain closed, the piston moves from TDC to BDC. The Crank rotation is from 0 to 180 deg.
- The high pressure and high temperature burnt gases force the piston to perform this stroke, called power stroke. This stroke is also known as expansion or working stroke.
- The engine produces mechanical work or power during this stroke.
- As the piston moves from TDC to BDC, the pressure of hot gases gradually decreases and volume increases. This is represented by curve 4-5 on the p-V diagram.
- Near the end of this stroke, the exhaust valve opens which will release the burnt gases to the atmosphere. This will suddenly bring the cylinder pressure to the atmospheric pressure.
- This drop of pressure at constant volume is represented by vertical line 5-2 on the p-V diagram.

4. EXHAUST STROKE -

- During this stroke, the exhaust valve opens and the inlet valve remains closed.
- The piston moves from BDC to TDC and during this motion piston pushes the exhaust gases (combustion products) out of the cylinder at constant pressure. The Crank rotation is from 180 to 360 deg.
- This process is shown on p-V diagram by horizontal line 2-1 in fig.
- At the end of the exhaust stroke, the exhaust valve is closed and inlet valve will open. Then there will be again a suction stroke and the same cycle will be repeated.

- 25. Compare the following: - 1) S.I. engine with C.I. engine.
2) 2-stroke with 4-stroke I.C. engine.**

❖ S.I. (PETROL) ENGINE AND C.I. (DIESEL) ENGINE

Parameters	S.I. Engine (Petrol Engine)	C.I. Engine (Diesel Engine)
Thermodynamic cycle	Works on Otto cycle (constant volume cycle)	Works on Diesel cycle (constant pressure cycle)
Fuel used	Petrol or Gasoline	Diesel
Supply of fuel	In carburettor, the fuel gets mixed with air, then this mixture is passed on to cylinder during suction stroke	Diesel is pressurized with the help of fuel pump and then injected into the engine cylinder by the fuel injector at the end of compression stroke
Compression ratio	Low (6 – 10)	High (14 – 20)
Charge drawn during the suction stroke	Air and petrol mixture	Only atmospheric air
Fuel ignition	Compressed charge is ignited by spark plug	Fuel is ignited by the heat of compressed air; no any external source is required
Engine speed	High (3000 rpm)	Low to medium (500 – 1500 rpm)
Thermal efficiency	Low (due to lower compression ratio)	High (Due to higher compression ratio)
Weight of the engine	Lighter	Heavier
Initial and maintenance costs	Less	More
Running costs	High due to higher cost of petrol	Low due to lower cost of diesel
Starting of engine	Easier starting even in cold weather conditions	Difficult to start in cold weather conditions

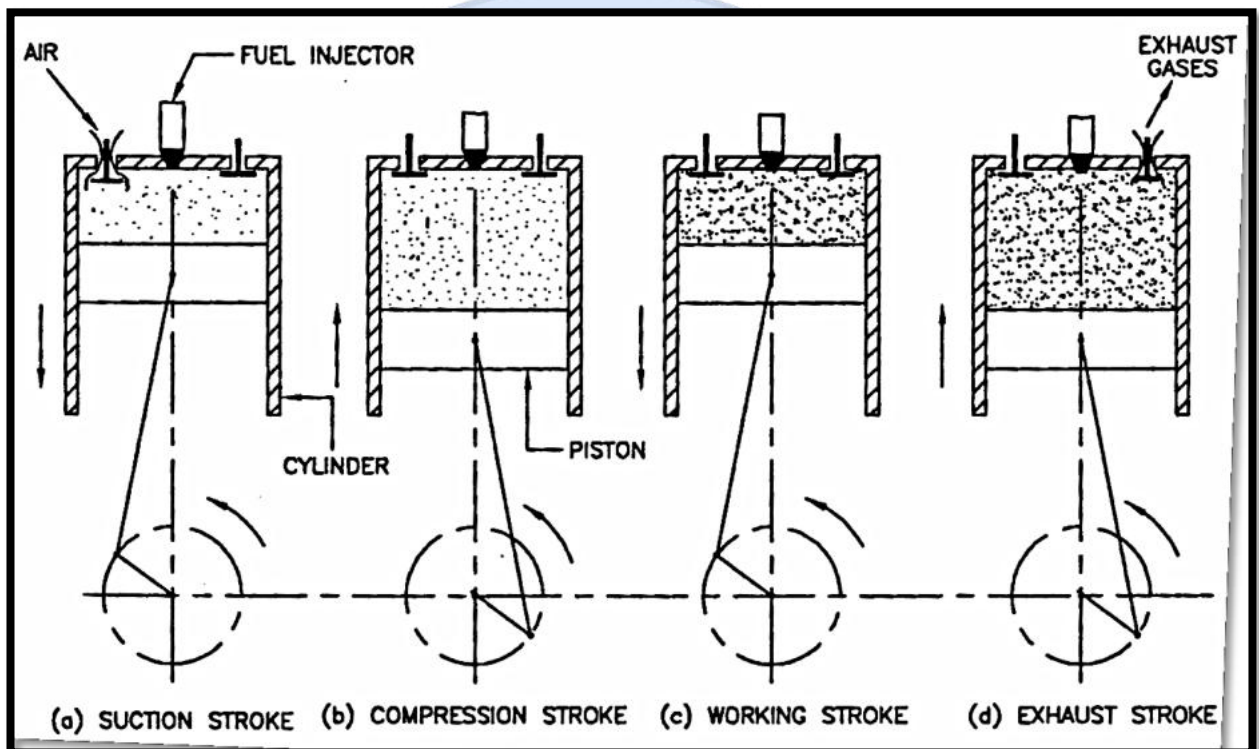
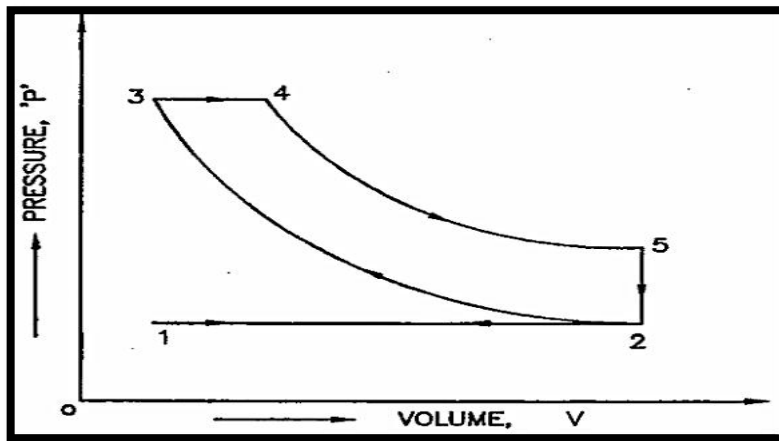
❖ 4-STROKE ENGINE AND 2-STROKE ENGINE

Parameters	4 – Stroke Engine	2 – Stroke Engine
Number of piston strokes per cycle	Four piston strokes required to complete a cycle	Only two piston strokes required to complete a cycle
Number of crank rotations per cycle	Two	Only one
Number of power strokes per minute	Equal to half the speed of engine crankshaft ($n = N/2$)	Equal to the speed of engine crankshaft ($n = N$)
Power developed	In every alternate revolution of crankshaft	In every revolution of the crankshaft
Flywheel	Heavy flywheel is required	Lighter flywheel is required
Size for the same power output	These engines are heavier, larger and require more space	These engines are lighter, more compact and require less space

Admission of charge	The charge is directly admitted into the engine cylinder during suction stroke	The charge is first admitted into the crankcase and then transferred to the engine cylinder
Valves / ports	The inlet and exhaust valves are required and are operated by the valve gear mechanism	In place of valves, Ports are present which are covered and uncovered by the motion of piston
Crankcase	Crankcase is not hermetically sealed	Crankcase is hermetically sealed because charge is admitted in it
Lubricating oil consumption	Less	More
Thermal efficiency	High because there is no mixing of fresh charge with exhaust gases	Low because there is mixing of fresh charge with exhaust gases
Mechanical efficiency	Less due to more number of moving parts	High due to less number of moving parts
Applications	These engines are used in high power applications where more space is available like cars, trucks, tractors, buses, etc.	These engines are used in low power applications where less space is available like scooters, bikes, etc

26. Explain with neat sketch working of 4-stroke diesel engine.

- The diesel engines work on the principle of Diesel cycle, also called constant pressure heat addition cycle.
- The basic construction of a four-stroke diesel engine is same as that of four stroke petrol engine, except instead of spark plug, a fuel injector is mounted in its place.
- A fuel pump supplies the fuel oil (Diesel) to the injector at higher pressure.
- The whole cycle is completed in four strokes, namely
 1. Suction stroke
 2. Compression stroke
 3. Working or Power or Expansion stroke
 4. Exhaust stroke
- All these strokes are represented as follows on PV diagram and the operating diagrams.



1. SUCTION STROKE –

- During this stroke, inlet valve remains open and exhaust valve remains closed, the pressure in the cylinder will be atmospheric.
- The piston moves from TDC to BDC position. The Crank rotation is from 0 to 180 deg.
- Air from the atmosphere is drawn into the cylinder as the piston moves. This stroke is represented by horizontal line 1-2 on p- V diagram shown in Fig.
- At the end of this stroke, the cylinder will be filled completely with air and inlet valve will be closed.

2. COMPRESSION STROKE –

- As shown in fig. during this stroke, both inlet valve and exhaust valve remain closed. The piston moves from BDC to TDC. The Crank rotation is from 180 to 360 deg.

- As this stroke is being performed, the air in the cylinder will be compressed, so pressure and temperature of air increases.
- The compression ratio of this engine is higher than petrol engine. Due to higher compression ratio, air will have attained a higher temperature than self-ignition temperature of the diesel fuel.
- Near the end of this stroke, the diesel fuel is injected by the Injector in the form of fine atomized spray into the cylinder.
- As the diesel fuel particles come in contact with high temperature air, it will ignite automatically. This is called auto-ignition or self-ignition.
- In this engine compressed air ignites the diesel fuel; this type of engine is also called as compression Ignition engine or C.I. engine.

3. EXPANSION OR POWER OR WORKING STROKE –

- During this stroke, both the inlet and exhaust valve remain closed. The piston moves from TDC to BDC. The Crank rotation is from 0 to 180 deg.
- The fuel injection starts nearly at the end of compression stroke, but the rate of fuel injection is such that combustion maintains constant pressure. It is shown on P-V diagram by horizontal line (3-4).
- The high temperature and high-pressure gases expand and push the piston downward. Thus, useful work is done during this stroke.
- As the piston moves from TDC to BDC, the pressure of hot gases gradually decreases and volume increases. This is represented by curve (4-5) on the P-V diagram shown in fig.
- As the piston reaches BDC position, the exhaust valve opens, the excited exhaust gases are removed from the cylinder and the pressure suddenly falls nearly to atmospheric pressure at a constant volume. It is shown by line (5-2) on P-V diagram.

4. EXHAUST STROKE –

- During this stroke, the exhaust valve remains opened and inlet valve is closed. The piston moves from BDC to TDC. The Crank rotation is from 180 to 360 deg.
- During this motion, piston pushes the exhaust gases (combustion products) out of cylinder at constant pressure.
- This process is shown on p-V diagram by horizontal line (2-1).
- Again, inlet valve opens and a new cycle starts.

27. The following data is available for 2-stroke diesel engine: Bore=10 cm, stroke=15 cm, engine speed=1000 RPM, Torque developed=58 N-m, $\eta_m=80\%$, indicated thermal efficiency=40%, Calorific value of fuel=44000 kJ/kg. Find: (a) Indicated Power (b) Mean effective Pressure & (c) Brake Specific Fuel Consumption.

• **Given Data:**

- Bore = $d = 10 \text{ cm} = 0.1 \text{ m}$
- Stroke = $L = 15 \text{ cm} = 0.15 \text{ m}$
- Engine speed = $N = 1000 \text{ rpm}$
- Torque developed = $T = 58 \text{ N-m}$
- Mechanical efficiency = $\eta_m = 80\% = 0.8$
- Indicated thermal efficiency = $\eta_{it} = 40\% = 0.4$
- Calorific value of fuel = $44000 \text{ kJ/kg} = 44 \times 10^6 \frac{\text{J}}{\text{kg}}$

• **To Find:**

- Indicated power; I.P
- Mean effective pressure; p_m
- Brake specific fuel consumption; BSFC

• **Calculation:**

- Brake power (B.P):

$$\bullet \quad B.P = \frac{2\pi NT}{60}$$

$$\bullet \quad B.P = \frac{2 \times \pi \times 1000 \times 58}{60}$$

$$\bullet \quad \mathbf{B.P. = 6.074 \text{ kW}}$$

- Indicated power (I.P):

$$\bullet \quad \eta_m = \frac{B.P.}{I.P.}$$

$$\bullet \quad I.P. = \frac{B.P.}{\eta_m}$$

$$\bullet \quad I.P = \frac{6.074 \times 10^3}{0.8}$$

$$\bullet \quad \mathbf{I.P. = 7.5925 \text{ kW}}$$

- Mean effective pressure (p_m):
 - $I.P = \frac{p_m L A N}{60}$ (Assume number of cylinder = 1)
 - $I.P = \frac{p_m L (\frac{\pi}{4} d^2) N}{60}$
 - $p_m = \frac{I.P \times 60 \times 4}{L \times \pi \times d^2 \times N}$
 - $p_m = \frac{7.5925 \times 10^3 \times 60 \times 4}{0.15 \times \pi \times 0.1^2 \times 1000}$
 - **$p_m = 3.87 \text{ bar}$**

- Brake Specific fuel consumption (BSFC):

- $BSFC = \frac{m_f (\text{in kg/hr})}{B.P (\text{in kW})} \dots (1)$
 - For m_f ,
 - $\eta_{it} = \frac{I.P.}{m_f \times C.V}$
 - $m_f = \frac{I.P.}{\eta_{it} \times C.V}$
 - $m_f = \frac{7.5925 \times 10^3}{0.4 \times 44 \times 10^6}$
 - $m_f = 1.5530 \text{ kg/hr}$
 - Put value of m_f in eq (1)
 - $BSFC = \frac{1.5530}{6.074}$
 - **$BSFC = 0.2556 \frac{\text{kg}}{\text{kW hr}}$**

28. The following readings were taken during the test of four stroke single cylinder petrol engine: Load on the brake drum = 50 kg. Diameter of brake drum = 1250 mm. Spring balance reading = 7 kg. Engine speed = 450 rpm. Fuel consumption = 4 kg/hr Calorific value of the fuel = 43000 kJ/kg. Calculate: (i) indicated thermal efficiency (ii) brake thermal efficiency. Assume mechanical efficiency as 70%.

- **Given Data:**

- Load on brake drum = $W = 50 \text{ kg}$
- Diameter of brake drum = $1250 \text{ mm} = 1.25 \text{ m}$
 $\therefore$ Effective Radius of brake drum = $R_b = 1.25 / 2 = 0.625 \text{ m}$
- Spring balance = $s = 7 \text{ kg}$
- Engine speed = $N = 450 \text{ rpm}$

- Fuel consumption = $m_f = 4 \frac{kg}{hr} = \frac{4}{3600} \frac{kg}{s}$
- Mechanical efficiency = $\eta_m = 70\% = 0.7$
- Calorific value of fuel = $43000 \text{ kJ/kg} = 43 \times 10^6 \frac{J}{kg}$

• **To Find:**

- Indicated thermal efficiency; η_{it}
- Brake thermal efficiency; η_{bt}

• **Calculation:**

- Brake thermal efficiency (η_{bt}):

- $\eta_{bt} = \frac{B.P.}{m_f \times C.V.} \dots (1)$

- For B.P. ,

- $B.P = \frac{2\pi NT}{60}$

- $B.P = \frac{2\pi N(W-s)R_p}{60}$

- $B.P = \frac{2 \times \pi \times 450 \times (50-7) \times 9.81 \times 0.625}{60} \quad (\because \text{Net brake load} = (W-s) \times g)$

- $B.P = 12.424 \text{ kW}$

- Put value of B.P in eq (1)

- $\eta_{bt} = \frac{12.424 \times 10^3}{\frac{4}{3600} \times 43 \times 10^6}$

- $\eta_{bt} = 26.00\%$

- $\eta_{bt} = 26\%$

- Indicated thermal efficiency (η_{it}):

- $\eta_{it} = \frac{I.P.}{m_f \times C.V.} \dots (2)$

- For I.P. ,

- $\eta_m = \frac{B.P.}{I.P.}$

- $I.P. = \frac{B.P.}{\eta_m}$

- $I.P = \frac{12.424 \times 10^3}{0.7}$

- $I.P. = 17.748 \text{ kW}$

- Put value of I.P in eq (2)

- $\eta_{it} = \frac{17.748 \times 10^3}{\frac{4}{3600} \times 43 \times 10^6}$
- $\eta_{it} = 37\%$

29. Draw line diagram of vapour compression refrigeration cycle and represent on P-h and T-S diagram and state function of individual components of vapour compression refrigeration system.

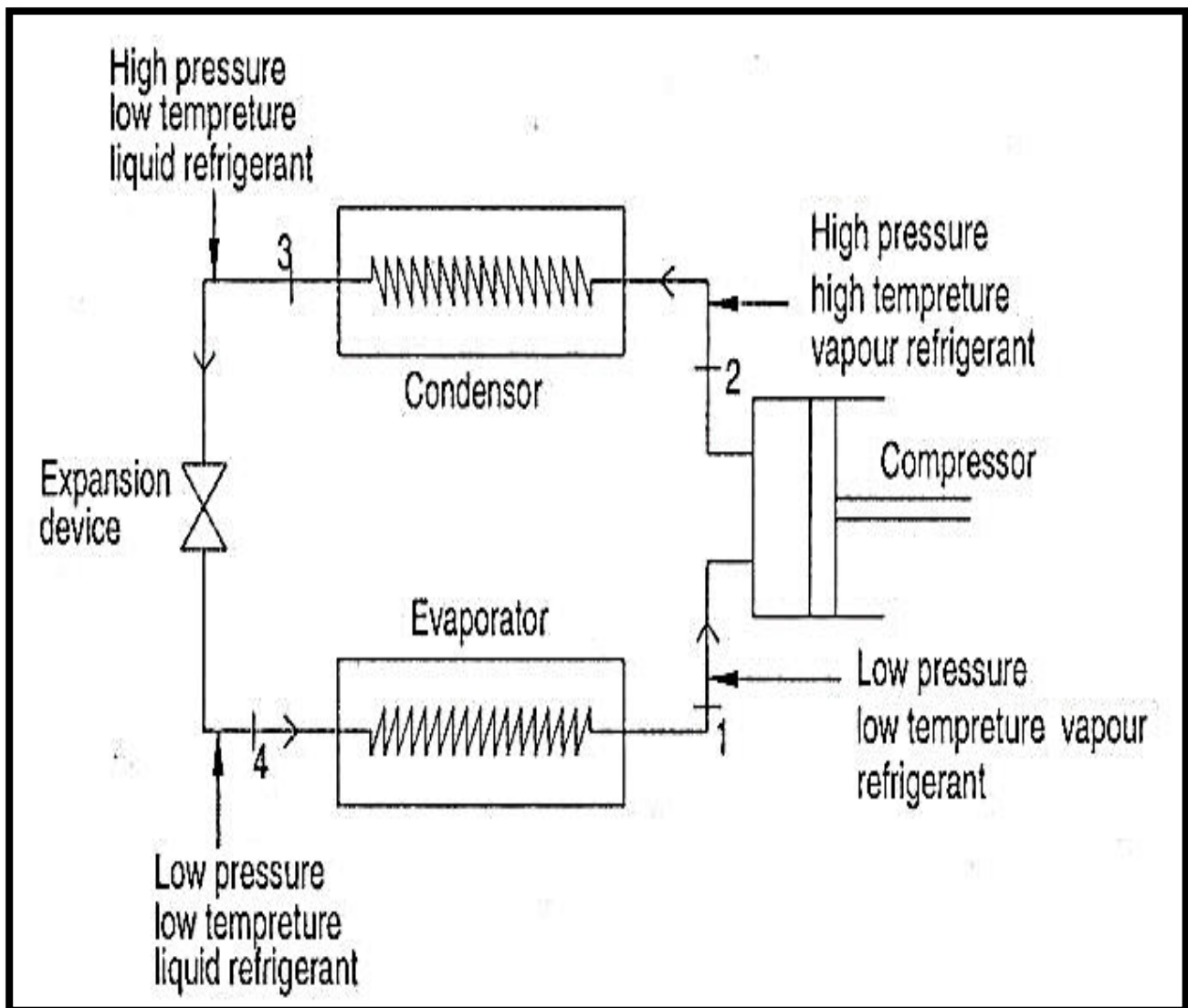


Figure – Line diagram for Vapour compression refrigeration (VCR) cycle.

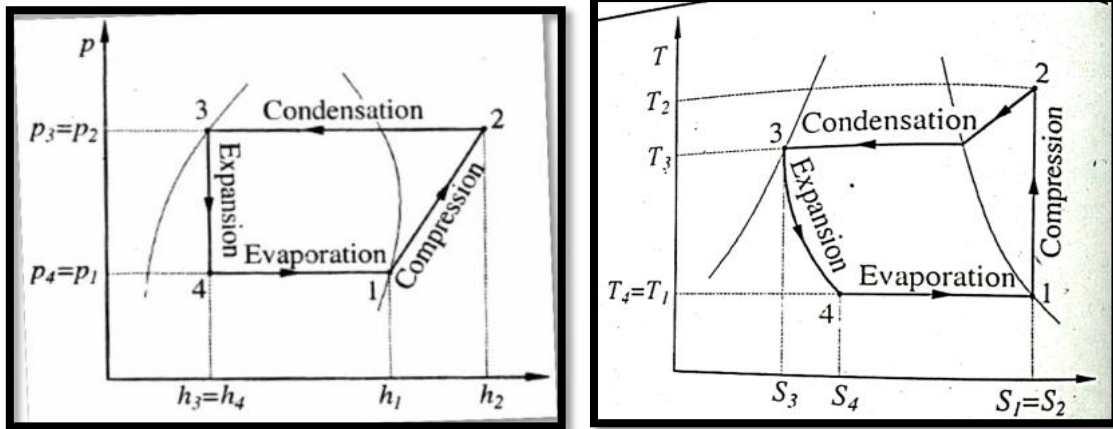


Figure – p-h and T-s diagram from Vapour compression refrigeration (VCR) cycle.

- The VCR system consist of (1) Evaporator (2) Compressor (3) Condenser and (4) Expansion device.
- In vapour compression refrigeration cycle, vapour is used as the refrigerant. The refrigerant is circulated in system where it alternately evaporates (liquid to vapour) and condenses (vapour to liquid); thus it undergoes a change of phase.
- During the evaporation process in the evaporator, the refrigerant absorbs the latent heat of evaporation from the space to be cooled.
- In the condenser, the refrigerant rejects heat to atmosphere.

• Function of main parts of Vapour compression refrigeration system.

❖ COMPRESSOR

- Function of compressor is to remove the vapour from the evaporator and increases its pressure and temperature up to it can be condensed in the condenser.
- Pressure of refrigerant coming from compressor should be such that the saturation temperature of vapour (corresponding to this pressure of vapour) is higher than the temperature of cooling medium in condenser.
- So that high pressure vapour can reject heat to cooling medium in the condenser.

❖ CONDENSER

- The function of condenser is to facilitate a heat transfer surface through which heat transfer takes place from the hot refrigerant vapour to the condensing medium. In domestic refrigerator condensing medium is atmospheric air.

❖ EXPANSION VALVE OR DEVICE

- The function of expansion valve is to meter the proper amount of liquid refrigerant and reduces pressure of liquid refrigerant entering the evaporator.
- Hence liquid will vaporize in the evaporator at the desired low temperature and absorb heat from the space.

❖ EVAPORATOR

- An evaporator provides a heat transfer surface through which low temperature liquid refrigerant can absorb latent heat from space and it vaporized.

30. What is brake? Describe an Internal expanding shoe brake with a neat sketch.

- A brake is a mechanical device which produces frictional resistance against the moving machine, in order to slow down or stop the motion of the machine.

Internal Expanding Shoe Brake.

- An internal expanding shoe brake consists of two shoes S_1 and S_2 as shown in figure.
- The outer surface of the shoes are covered with friction material.
- Each shoe is pivoted at one end about a fixed fulcrum O_1 and O_2 and other end rests against the cam.
- When the cam is operated, the shoes are pushed outwards against the brake drum.
- The friction between the shoes and the drum produces the braking torque. Thus reduces the speed of the rotating drum.
- This type of brake is commonly used in motor cars and light trucks.

